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WHEN DID GROWTH BEGIN? NEW ESTIMATES OF PRODUCTIVITY GROWTH IN ENGLAND FROM 1250 TO 1870*

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We estimate productivity growth in England from 1250 to 1870. Real wages over this period were heavily influenced by plague-induced swings in the population. Our estimates account for these Malthusian dynamics. We find that productivity growth was zero before 1600. Productivity growth began in 1600—almost a century before the Glorious Revolution. Thus, the onset of productivity growth preceded the bourgeois institutional reforms of seventeenth-century England. We estimate productivity growth of 2% per decade between 1600 and 1800, increasing to 5% per decade between 1810 and 1860. Much of the increase in output growth during the Industrial Revolution is explained by structural change—the falling importance of land in production—rather than faster productivity growth. Stagnant real wages in the eighteenth and early nineteenth centuries—Engels’ Pause—is explained by rapid population growth putting downward pressure on real wages. Yet feedback from population growth to real wages is sufficiently weak to permit sustained deviations from the “iron law of wages” prior to the Industrial Revolution. *JEL codes:* N13, O40, J10.

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I. INTRODUCTION

When did economic growth begin? A traditional view holds that economic growth began with the Industrial Revolution around 1800. Recent work has challenged this view, pushing back the date of the onset of growth. [Crafts \(1983, 1985\)](#) and [Harley \(1982\)](#) revised downward previous estimates of growth in Britain during the Industrial Revolution. These new estimates indicate that British output per capita was larger by the mid-eighteenth century than was previously thought, implying that substantial growth must have occurred at an earlier date (see also [Crafts and Harley 1992](#)). [Acemoglu, Johnson, and Robinson \(2005\)](#) argue that a first Great Divergence occurred starting around 1500 with Western Europe growing apart from other areas of the world after the discovery of the Americas and the sea route to India. They support this view with data on urbanization rates. [Broadberry et al. \(2015\)](#) argue that growth began even earlier than this. They present new estimates of GDP per person for Britain back to 1270. These data show slow but steady growth in GDP per person from the beginning of their sample. Finally, [Kremer \(1993\)](#) uses world population estimates to argue for positive but glacially slow growth for hundreds of thousands of years.

An important facet of the debate about when growth began is when productivity growth began. We contribute to this debate by constructing a new series for productivity growth (total factor productivity, TFP) in England back to 1250. [Figure I](#) plots our new productivity series (solid black line). Our main finding is that productivity growth in England began in 1600. Between 1250 and 1590, we estimate that productivity growth was zero.¹ We estimate productivity growth of about 2% per decade between 1600 and 1800. Productivity growth then increased to 5% per decade between 1810 and 1860. We attribute much of the increase in output growth during the Industrial Revolution to structural change—a fall in the importance of land in production—rather than to an increase in productivity growth.

Our results help distinguish between different theories of why growth began. They suggest that researchers should focus

1. The positive but glacially slow productivity growth rate implied by [Kremer's \(1993\)](#) population data for the period 1200 to 1500 lies within our credible set.

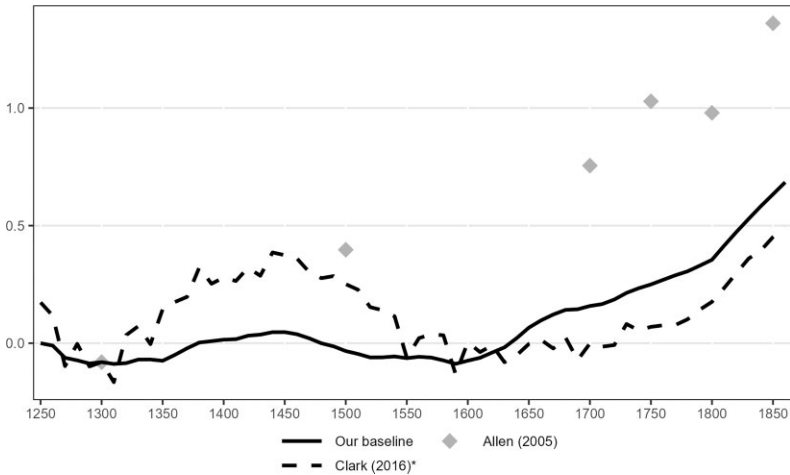


FIGURE I

Estimates of Productivity in England

Each series is the natural log of productivity. The series denoted by “Clark (2016)*” is the series from Clark (2016) extended to 1860. We received this series from Clark in private correspondence. Clark’s series estimates TFP for the entire economy based on a dual approach. Allen’s (2005) estimates are for TFP in the agricultural sector using a primal approach. Our preferred productivity series is normalized to zero in 1250. The other two series are normalized to match our preferred series in 1300.

on developments proximate to the sixteenth and seventeenth centuries. An important debate regarding the onset of growth centers on the role of institutional change. Our results help sharpen this debate. We find that productivity growth began almost a century before the Glorious Revolution and well before the English Civil War. While the institutional changes associated with these events may have been important for subsequent growth, researchers must look to earlier events for the seeds of modern growth. Plausible candidates include the Reformation, the decline of feudalism, the rise of the yeoman, movable type printing and the associated increase in literacy, and expansion of international trade. We discuss these in more detail below.

The most comprehensive existing productivity series for England was constructed by Clark (2010, 2016). Clark estimated changes in TFP for the entire English economy from 1209 onward using the “dual approach”—that is, as a weighted average of changes in real factor prices (e.g., Hsieh 2002). Figure I plots

Clark's series over our sample period (broken black line).² A striking feature of this series is that it implies that productivity in England was no higher in the mid-nineteenth century than in the fifteenth century. This result does not line up well with other existing (less comprehensive) measures of productivity in England or with less formal assessments of the English economy. For example, Allen (2005) estimates that TFP in agriculture was 162% higher in 1850 than in 1500 (gray diamonds in Figure I).³ Clark himself commented that if the fluctuations in his series are not measurement error "they imply quite inexplicable fluctuations in the performance of the preindustrial economy."

Our conclusions about productivity in England are quite different from those of Clark (2010, 2016). According to our estimates, productivity in England was 95% higher in 1850 than in 1500 rather than being essentially unchanged. We also estimate smaller fluctuations in productivity prior to 1600. In particular, our productivity series falls much less between 1450 and 1600. These substantial differences arise from differences in the data and methodology we use. We take the labor demand curve as our starting point and estimate changes in productivity as shifts in the labor demand curve. This means that the key data series that inform our estimates are real wages and population. These are arguably among the best measured series of all economic time series over our long sample period.

Our approach is best understood by considering Figure II. This figure presents a scatter plot of the log of real wages in England (y-axis) against the log of the population in England (x-axis). From 1250 to 1300, the population of England increased and real wages decreased. The period from 1300 to 1450 was a period of frequent plagues—the most famous being the Black Death of 1348. Over this period, the population of England fell by a factor of two and real wages rose substantially. From 1450 to 1600, the population recovered and real wages fell. In 1630, the English

2. Clark (2016) published an update of his better-known 2010 series for the shorter time period 1250–1600. The series we plot in Figure I is Clark's 2016 series extended to 1860. We received this series from Clark by private correspondence. The 2016 series differs from the 2010 series prior to 1600 due to a new land rent series and because Clark corrected an important error in the 2010 series. We discuss this in more detail in Online Appendix H.

3. Allen (2005) employs the familiar "primal approach" of Solow (1957), that is, subtracts a weighted average of growth in factor inputs from output growth, but is only able to do this for agriculture and for a few years.

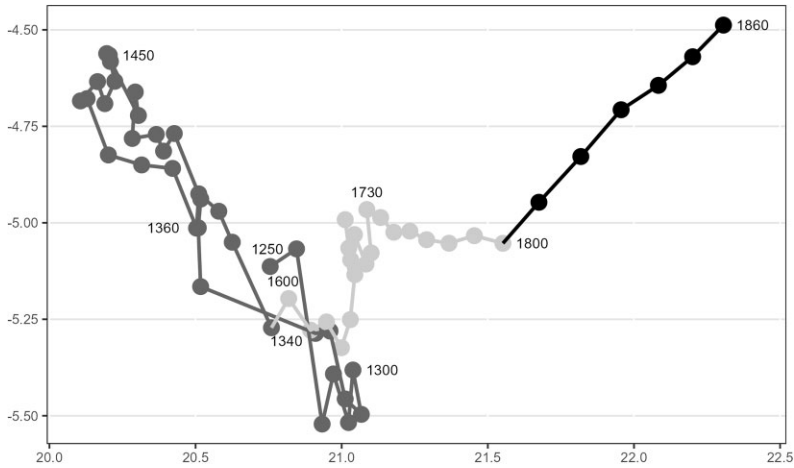


FIGURE II
Real Wages and Population

The figure presents a scatter plot of the log of real wages in England against the log of the population in England over the period 1250–1860. The data on real wages are from [Clark \(2010\)](#). Estimates of the population are based on our calculations (baseline case).

economy was back to almost exactly the same point it was at in 1300.

One way to explain these dynamics between 1300 and 1630 is as movements along a stable labor demand curve with no change in productivity. Had productivity grown between 1300 and 1630, the economy could not have returned to essentially the same point in 1630 as it was in 1300 since the labor demand curve would have shifted up and to the right over the intervening period. Then in the seventeenth century, something important seems to change. The points start moving off the prior labor demand curve. Specifically, they start moving up and to the right relative to the earlier curve. This suggests that productivity started growing in the seventeenth century in England.

The basic idea behind our approach is to estimate a labor demand curve for England and then back out productivity growth as shifts in this labor demand curve. To get a better sense for how this approach works, consider the following simple labor demand curve for a premodern economy:

$$W_t = (1 - \alpha)A_t \left(\frac{Z}{L_t} \right)^\alpha,$$

where W_t denotes real wages, A_t denotes productivity (TFP), Z denotes land (which is fixed), and L_t denotes labor. We consider more general models later in the article, but the basic challenge we face can be grasped using this simple model. If we take logs, this equation becomes

$$w_t = \phi - \alpha l_t + a_t,$$

where lowercase letters denote logs of uppercase letters. Armed with data on real wages, the population, and an estimate of the slope of the labor demand curve α , one can back out estimates of productivity a_t .

We consider two approaches to estimating the slope of the labor demand curve α . The first is to use the Black Death as a large exogenous shock to the population. In this case, we estimate α simply from data on real wages and population before and after the Black Death. Our second method is to structurally estimate a Malthusian model. In this case, we are modeling the endogenous response of the population to changes in the real wage. These two methods yield similar results.

Because our analysis extends into the early industrial era, we must confront the fact that the importance of land as a factor of production fell rapidly with the spread of steam power, which meant that the production of energy was no longer land intensive (Wrigley 2010). To capture this crucial development, we allow the output elasticity of land, capital, and labor to change over time after the onset of the Industrial Revolution. We use data on land rents after 1760 to pin down how rapidly the importance of land in production fell. The modest increase in land rents that we observe in the face of explosive growth in labor and capital after 1760 leads us to estimate a rapidly falling importance of land in production.

The fact that we allow for this structural transformation implies that the standard way of measuring productivity (a multiplicative A_t in front of a function $F(L_t, K_t, \dots)$) is no longer valid. Following Caves, Christensen, and Diewert (1982) we derive a Malmquist productivity index (Malmquist 1953) for our setting. The Malmquist index reduces to A_t in the familiar setting of constant factor elasticities, but remains valid even when the structure of the production function is changing.

Allowing the importance of land in production to fall after the start of the Industrial Revolution has important implications for our estimates of productivity. If we don't allow for this change, we

estimate a much larger break in productivity in 1810. Productivity growth is, of course, just a measure of our ignorance. Modeling the shift of production away from land-intensive technology allows us to explain a larger part of growth during the Industrial Revolution, leaving less for the residual.

Our estimates shed light on the lack of real wage growth during the latter part of the eighteenth century, sometimes referred to as Engels' Pause ([Engels 1845](#); [Allen 2009b](#)). Our Malthusian model implies that during this period real wages were held back by very rapid increases in the population, which in a Malthusian world put downward pressure on the marginal product of labor. This explanation contrasts with the common idea that the absence of real wage growth during this period resulted from the lion's share of the fruits of technical change going to capital as opposed to labor. This idea has received attention in the modern context in relation to the development of automation and artificial intelligence ([Acemoglu and Restrepo 2019a](#)).

In addition to estimates of productivity, our methodology yields estimates of the speed of Malthusian population dynamics in premodern England. Our estimates imply that these population dynamics were very slow: a doubling of real incomes led to an increase in population growth that was only about 3 to 6 percentage points per decade. Together with our other estimates, this implies that the half-life of a plague-induced drop in the population was more than 100 years before the onset of the Industrial Revolution. As the importance of land in production fell after 1760, the Malthusian population dynamics became even slower and weaker. By 1860, our estimate of the half-life of a population shock have risen to several hundred years. Earlier estimates of the speed of Malthusian population dynamics in England also indicate that they were slow. For example, [Lee and Anderson \(2002\)](#) find a half-life of 107 years, while [Crafts and Mills \(2009\)](#) find a half-life of 431 years. [Chaney and Hornbeck \(2016\)](#) document very slow population dynamics in Valencia after the expulsion of the Moriscos in 1609.

The weakness of the Malthusian population dynamics we estimate imply that our model is consistent with sustained deviations from "the iron law of wages" (i.e., that wages in a Malthusian economy are stuck at subsistence). Modest productivity growth over a few centuries can temporarily overwhelm the Malthusian population dynamics in our model and result in sustained periods of real wages several times higher than at other times. Our model

can therefore make sense of episodes that historians sometimes refer to as “golden ages” or “efflorescences” (Goldstone 2002). Once productivity growth falters, real wages will slowly fall back to a lower level. But this will take several centuries.

We are not the first to plot a figure like Figure II and argue that it has implications about the evolution of productivity in England. We formalize this intuitive idea and assess what exactly it implies about productivity. Clark (2005, 2007a) discusses informally how shifts in the labor demand curve of a Malthusian model can be informative about the timing of the onset of economic growth. The existing papers most closely related to ours from a methodological point of view are Lee and Anderson (2002) and Crafts and Mills (2009). These papers structurally estimate a Malthusian model of the English economy, as we do. Relative to these publications, we extend the sample period back in time considerably (theirs starts in 1540, while ours start in 1250). This allows us to assess when growth began. We also estimate α differently, incorporate capital, and allow the importance of land to change after the onset of the Industrial Revolution, among other differences.

The timing of the onset of productivity growth that we estimate lines up well with recent estimates of the onset of structural transformation in Wallis, Colson, and Chilosì (2018). They estimate that the share of workers in agriculture began a long and large fall around 1600 after having been stable in the sixteenth century. Making use of data on GDP and population from Broadberry et al. (2015), they also estimate that labor productivity in agriculture, industry, and services began to rise around 1600.

This article is also related to the literature in macroeconomics on the transition from preindustrial stagnation to modern growth—often referred to as the transition “from Malthus to Solow.” Important papers in this literature include Galor and Weil (2000), Jones (2001), and Hansen and Prescott (2002). Relative to these works, our study is more empirical. We contribute detailed estimates of the evolution of productivity, while the other papers propose theories of how productivity growth rose. Our work is also related to recent work by Hansen, Ohanian, and Ozturk (2020).

The article proceeds as follows. Section II presents a simple estimate of productivity growth in England with α estimates from the Black Death. Section III presents our full Malthusian model

of the economy. [Section IV](#) presents our results on productivity based on the full model. [Section V](#) presents our results on the strength of the Malthusian population force. [Section VI](#) presents our estimates of the population. [Section VII](#) concludes.

II. A SIMPLE ESTIMATE OF PRODUCTIVITY GROWTH IN ENGLAND

We begin by presenting a very simple estimate of productivity growth in England. Later sections develop a number of extensions and refinements to the basic approach adopted here. We model time as discrete and denote it by a subscript t . Because we use decadal data throughout the article, each time period in the model is meant to represent a decade. Our sample period is from 1250 to 1860. All the data we use are decadal averages. In our figures, a data point listed as 1640 refers to the decadal average from 1640 to 1649. We sometimes refer to a variable at a point in time (say, 1640) when we mean the decadal average for that decade. In other words, we use 1640 and “the 1640s” interchangeably.

Consider an economy where output is produced with land and labor according to the following production function:

$$Y_t = A_t Z^\alpha L_t^{1-\alpha},$$

where Y_t denotes output, Z denotes land, L_t denotes labor, and A_t denotes productivity (TFP). We model the quantity of land as being fixed. Literally speaking, the stock of land in England has always been fixed. However, the way land has been used has changed over time. The sharp fall in population after the Black Death led land to be converted to pastoral farming (which is land intensive). Growth in the population later in our sample period resulted in more land being used for arable farming (less land intensive). A growing population implied that people were forced to use lower and lower quality land to grow crops and farm each hectare of land more intensively. This meant that for a given level of productivity, the marginal product of labor was decreasing in the size of the population. Our production function with Z being fixed captures this fact.⁴

Land use also changed due to improvements such as draining swamps, clearing fields, and cutting down forests. Later we

4. This is entirely analogous to the standard way the use of capital in production is usually modeled. Consider an ice cream maker who has one ice cream machine and one worker. If they hire more workers, the machine will be used more intensively and the marginal product of labor will fall.

consider a model with capital. Investments in land improvement are captured as additions to the capital stock. Finally, institutional change resulted in changes in land use. Two important examples are the confiscation and sale of monastic land by Henry VIII (Heldring, Robinson, and Vollmer 2021) and enclosures (Allen 1992). The effects of these institutional changes will show up as changes in productivity in our analysis.

We assume that producers hire workers in a competitive labor market taking wages as given. Producer optimization then gives rise to the following labor demand curve:

$$W_t = (1 - \alpha)A_t Z^\alpha L_t^{-\alpha},$$

where W_t denotes the real daily wage. Taking logs of this equation yields

$$(1) \quad w_t = \phi + a_t - \alpha l_t,$$

where lowercase letters denote logs of uppercase letters and $\phi = \log(1 - \alpha) + \alpha \log Z$. Assuming that labor is paid its marginal product in preindustrial England is a strong assumption that we discuss in greater detail below.

To estimate changes in productivity using equation (1), we need data on real wages, data on labor supply, and an estimate of the slope of the labor demand curve α . We discuss these in turn.

II.A. Data on Real Wages and the Population

Our baseline measure of real wages in England is Clark's (2010) series for unskilled building workers. Figure III plots this series. The main features of the series are a large and sustained rise between 1300 and 1450, a large and sustained fall between 1450 and 1600, some recovery over the seventeenth century, stagnation during the eighteenth century, and finally a sharp increase after 1800. Online Appendix Figure A.1 compares this series with several other series for real wages in England. This comparison shows that the real wage series we use is quite similar to Clark's real wage series for farmers and for craftsmen. We have redone our analysis with these series and discuss this analysis in Section IV.C.⁵

5. Much controversy has centered on the behavior of real wages in England between 1770 and 1850. This debate revolves around the extent to which laborers shared in the benefits of early industrialization (Feinstein 1998; Clark 2005; Allen 2007, 2009b). In Online Appendix Figure A.1, we also plot Allen's (2007) wage se-



FIGURE III

Real Wages in England, 1250–1860

The figure presents estimates of the real wages of unskilled building workers in England from [Clark \(2010\)](#).

Our baseline assumption is that labor supply is proportional to the population of England. [Figure IV](#) presents the population data we use. For the period from 1540 onward, we use population estimates from [Wrigley et al. \(1997\)](#), which in turn build on the seminal work of [Wrigley and Schofield \(1981\)](#). Sources for population data before 1540 are less extensive. [Clark \(2007b\)](#) uses unbalanced panel data on the population of villages and manors from manorial records and penny tithing payments to construct estimates of the population prior to 1540. We build on Clark's work to construct an estimate of the population before 1540.

We cannot directly use Clark's pre-1540 population series because the method for constructing his series involves making assumptions about the evolution of productivity.⁶ Since we aim to use the population series to make inference about the evolution of productivity in England, we cannot use a population series that already embeds assumptions about productivity growth. However, as an intermediate input into constructing his pre-1540

ries (which starts in 1770). The figure shows that the differences discussed in the prior literature are modest from our perspective and therefore do not materially affect our analysis.

6. [Online Appendix B](#) discusses Clark's method in more detail.

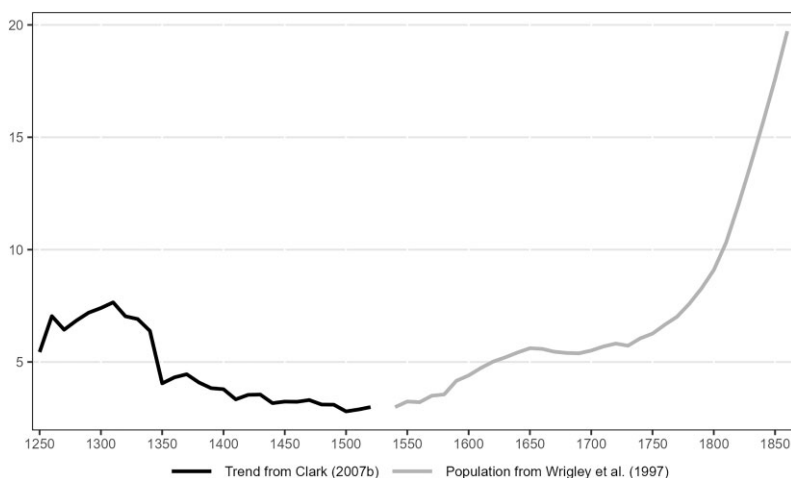


FIGURE IV

Population Data for England, 1250–1860

The figure presents population estimates for England for 1540–1860 from [Wrigley et al. \(1997\)](#) (gray line) and the population trend estimates by [Clark \(2007b\)](#) for the period 1250–1520 (black line). The black line is normalized for visual convenience such that its last point is equal to the first point of the gray line.

population series, Clark estimates a regression of his village- and manor-level population data on time and village/manor fixed effects. Clark refers to the time effects from this regression as a population trend. We plot this population trend in [Figure IV](#) (normalized for visual convenience). We base our estimates of the population of England before 1540 on this population trend series. In [Section IV.C](#), we discuss how this series compares with (lower frequency) population data reported in [Broadberry et al. \(2015\)](#).

We assume that the true population is measured with error. Specifically, we assume that $n_t = \psi + \tilde{n}_t + \iota_t^n$, where n_t denotes the true unobserved population, $\tilde{n}_t$ denotes our observed population series (Clark's population trend series prior to 1530 and the population series from [Wrigley et al. 1997](#) after 1530), $\iota_t^n \sim t_{\nu_n}(0, \sigma_n^2)$ denotes measurement error, and ψ denotes a normalization constant. We normalize ψ to zero after 1530 and estimate its value for the pre-1530 Clark series. We allow for a structural break in the variance of the measurement error σ_n^2 in 1540. Finally, population data are missing for 1530. We view it as unobserved and estimate its value.

II.B. Identification and Estimation

We estimate the slope of the labor demand curve— α in [equation \(1\)](#)—under the assumption that the Black Death is an exogenous shock to the population. Specifically, we estimate α as the ratio of the change in real wages and the change in the population between 1340 and 1360. This is the slope between the point for 1340 and the point for 1360 in [Figure II](#).⁷ The identifying assumption is that the Black Death dominates the change in wages and population between 1340 and 1360. We choose 1360 as opposed to 1350 as our post-Black Death point due to attempts by the lords in England to keep wages low in the immediate aftermath of the Black Death. In 1351, the Statute of Laborers was passed, which set a maximum wage equal to the prevailing wage prior to the Black Death. These attempts may have had some effect on wages during the 1350s, but wages responded strongly in the 1360s.

This procedure yields a value of $\alpha = 0.70$, which lines up well with the overall slope of the points in [Figure II](#) before the seventeenth century. This suggests that much of the variation in the population and real wages of England before the seventeenth century was driven by labor supply shocks such as plagues and other disease. With a Cobb-Douglas production function, α is also the land share of production. However, in [Online Appendix C](#), we show that the relationship between the land share of production and the slope of the labor demand curve is sensitive to the elasticity of substitution between land and labor. A constant elasticity of substitution production function implies that the land share is $\sigma\alpha$, where σ is the elasticity of substitution between land and labor. Assuming an elasticity of substitution of 0.66 ([Boppart et al. 2023](#)) implies a land share of $0.66 \times 0.70 = 0.46$. More generally, additional parameters are needed to go from the slope of the labor demand curve to factor shares.

Armed with data on w_t and l_t and an estimate of α , we can “back out” values for log productivity, a_t , from [equation \(1\)](#) (up to a constant). Because we are primarily interested in persistent changes in productivity, we filter out high-frequency variation in a_t by assuming that a_t is made up of a permanent and transi-

7. [Figure II](#) adjusts for measurement error. To estimate α , we use the raw data for 1340 and 1360. Our estimate of α is therefore only approximately the slope between the point for 1340 and the point for 1360 in [Figure II](#).

tory component. We assume the permanent component follows a random walk process with drift μ :

$$(2) \quad \tilde{a}_t = \mu + \tilde{a}_{t-1} + \epsilon_{1t}.$$

The average growth rate of productivity is given by the parameter μ . To capture changes in long-run growth, we allow for structural breaks in μ . Recall that changes in productivity correspond to shifts in the labor demand curve. We allow for a break in 1810 associated with the Industrial Revolution. But [Figure II](#) suggests that the labor demand curve began shifting out earlier. As a consequence, we allow for one additional break earlier in the sample. This allows for the possibility that there may have been a break in average growth before the Industrial Revolution. We consider a range of possible dates for the first break between 1550 and 1800.⁸ The method we use to pin down the timing of this earlier break and to estimate the model more generally is described in detail in [Section III](#).

II.C. When Did Growth Begin?

[Figure V](#) plots our estimates of the probability that a pre-Industrial Revolution break in productivity growth occurred at different dates between 1550 and 1800. The probability of a break spikes in 1600. It stays high in 1610 and 1620 and then falls off. The probability is very low prior to 1590 and also low after 1650. The probability of a break occurring before 1640—that is, before the English Civil War—is estimated to be 59%. The probability of a break occurring before 1680—that is, before the Glorious Revolution—is 73%. For expositional simplicity, we date the break in 1600. Results from our more complex models in [Sections III](#) and [IV](#) yield 1600 as the most likely break date (see [Online Appendix Figure A.2](#)).

[Table I](#) presents our estimates of the average growth rate of productivity μ for the three regimes over our sample. We estimate that average productivity growth prior to 1600 was zero. [Kremer \(1993\)](#) used data on the growth rate of the world population to argue that growth has been nonzero and increasing for

8. The fact that we allow for permanent breaks in productivity growth implies that productivity growth has a unit root component. This allows our model to match the fact that the population is integrated of order two in our sample, which has been emphasized in prior work on this topic ([Bailey and Chambers 1993](#); [Crafts and Mills 2009](#)).

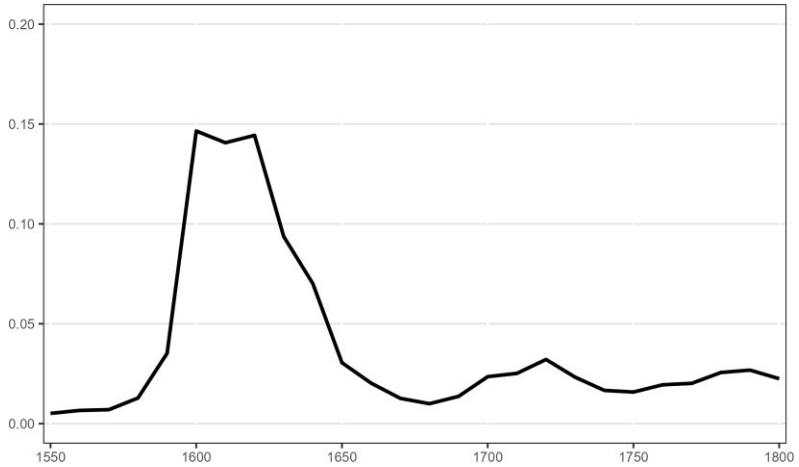


FIGURE V
Probability of Different Productivity Growth Break Dates

The figure plots our estimate of the probability that a structural break occurred in the parameters μ , σ_1 , and σ_2 in different decades between 1550 and 1800.

TABLE I
PRODUCTIVITY GROWTH

	Mean	Std. dev.	2.5%	97.5%
$\mu_{a,1}$	0.00	0.01	−0.01	0.02
$\mu_{a,2}$	0.04	0.02	0.02	0.10
$\mu_{a,3}$	0.19	0.01	0.17	0.22

Notes. The table presents the mean, standard deviation, 2.5% quantile, and 97.5% quantile of the posterior distribution we estimate for average productivity growth μ in the three regimes, using the simple procedure described in Section II. See Online Appendix Table A.1 for the posterior distribution of $\sigma_{\epsilon_{1,t}}$ and $\sigma_{\epsilon_{2,t}}$.

many millennia. The world population estimates he used indicate that world population growth from 1200 to 1500 was 0.6% per decade. In our Malthusian model (as well as Kremer’s model), steady-state productivity growth is α times steady-state population growth. Using our estimate of α discussed above, this suggests that growth in productivity was 0.4% per decade over the period 1200 to 1500, that is, positive but glacial. This slow growth rate is well within the credible set of our pre-1600 estimate of μ .

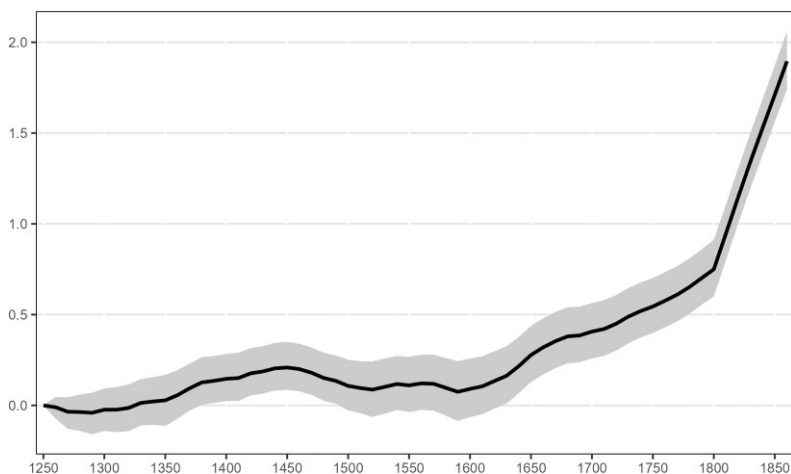


FIGURE VI

Permanent Component of Productivity

The figure plots our estimates of the evolution of the permanent component of productivity $\tilde{a}_t$ over our sample period (natural log relative to its value in 1250). The black line is the mean of the posterior for each period, and the gray shaded area is the 90% central posterior interval.

Our results indicate that sustained productivity growth began in 1600 (or around that time). We estimate average productivity growth of 4% per decade over the period 1600 to 1810. In the early nineteenth century, productivity growth accelerated sharply to 19% per decade. We conclude from these estimates that the period from 1600 to 1810 was a time of transition in England from an era of total stagnation to an era of modern economic growth. We refine these estimates in [Sections III](#) and [IV](#).

[Figure VI](#) presents our initial estimate of the time series evolution of the permanent component of productivity. These estimates indicate that the level of productivity in England was very similar in 1600 to what it had been in the late thirteenth century. In the intervening period, productivity fluctuated a slight bit. After 1600, productivity began a sustained increase, which accelerated sharply in 1810. As in most models in macroeconomics, productivity growth is a residual. In later sections, we consider more complex models with capital and in which we allow the elasticity parameters in the production function to change over time. In these cases, we attribute a substantial fraction of the post-1800

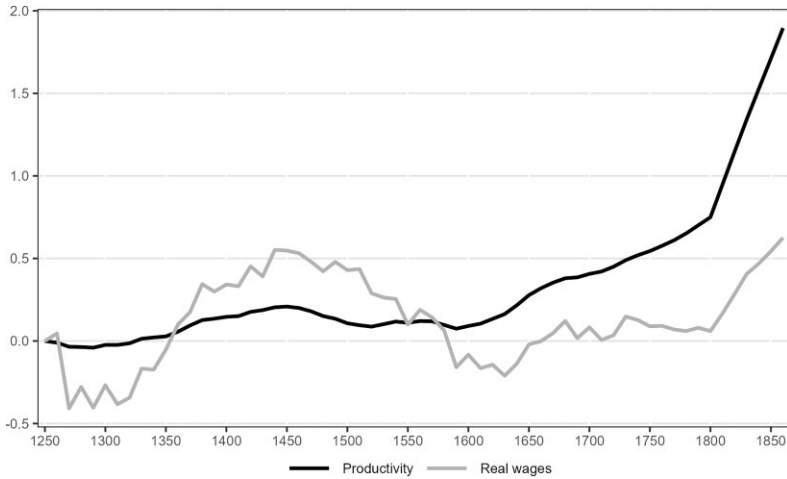


FIGURE VII

Productivity and Real Wages

The figure plots our estimates of the evolution of the permanent component of productivity $\tilde{a}_t$ along with the real wage series we use.

productivity growth in [Figure VI](#) to capital deepening and the falling importance of land in production.⁹

II.D. Real Wages, Productivity, and Engels' Pause

[Figure VII](#) compares our estimate of productivity with the data we use on real wages. This figure illustrates well the importance of accounting for Malthusian population forces when estimating productivity in the preindustrial era. Our analysis implies that the large changes in real wages before 1600 are explained almost entirely by changes in the population and almost not at all by changes in productivity. During this period, the economy moved up and down a relatively stable labor demand curve as suggested by [Figure II](#). First, plagues reduced the population and this increased wages. Then, the population recovered from these

9. [Online Appendix Table A.1](#) presents estimates of the standard deviation of the permanent and transitory productivity shocks. These vary very little across the three regimes. The standard deviation of the permanent productivity shocks is 0.03 prior to 1600 and 0.02 after 1600. The standard deviation of the transitory productivity shocks is 0.05 prior to 1600 and 0.04 after 1600.

plagues and real wages fell. As a result, changes in productivity were very substantially muted relative to changes in real wages.

Real wages are also a poor guide to changes in productivity for the period from 1600 to 1860. Over this period, however, the pattern is reversed: productivity moved substantially more than real wages. Productivity started to grow after 1600. But this induced rapid growth in the population. Since the labor demand curve was downward sloping due to land being a fixed factor, growth in the population put downward pressure on wages. Over this period, real wages, therefore, lagged behind growth in productivity.

Our analysis provides a simple explanation for Engels' Pause, the fact that real wages did not rise appreciably during the early decades of industrialization in England. We see in [Figure VII](#) that over 1730 and 1800, real wages in England fell slightly despite substantial productivity growth. (The real wage series of [Feinstein \(1998\)](#) and [Allen \(2007\)](#)—which differ somewhat from Clark's real wage series—extend the pause a few decades into the nineteenth century.) One explanation for this fact—famously articulated by [Engels \(1845\)](#)—is that the gains from capitalism overwhelmingly accrue to capitalists as opposed to laborers. Our analysis suggests an alternative Malthusian explanation: rapid growth in the population put downward pressure on the marginal product of labor and thus reduced the growth in wages relative to productivity.¹⁰

One reason we chose to use the wage series for unskilled building workers in our analysis is to capture a part of the labor market that involved relatively voluntary labor and therefore a better measure of the marginal product of labor than, say, the

10. [Allen \(2009b\)](#) comes to a similar conclusion using quite different data and methods. He also concludes that population growth was a crucial contributor to stagnant real wages in England during Engels' Pause: "population growth was a necessary condition for stationary real wages: Engels' pause looks like Malthus' dismal science come true" (430). Allen's model is more complex than the simple model we analyze in this section. In his model, slow accumulation of capital and a low elasticity of substitution between capital and labor also contribute to low growth in real wages during Engels' Pause. We extend our model to include capital in [Section III](#). In Allen's model, growth in capital eventually catches up to growth in the population leading real wage growth to pick up. Our later analysis places a greater emphasis on reductions in the importance of land in production in eliminating the Malthusian character of growth over the course of the nineteenth century.

work of the villein in the countryside. Nevertheless, our assumption of a competitive labor demand curve is clearly a simplification, and it is important to consider how our results might change were we to weaken this assumption. A constant wage markdown due to monopsony power or coercion by employers would not affect our results. It would only affect the estimate of the constant ϕ in [equation \(1\)](#). A wage markdown that was getting smaller over time as the labor market became more free and competitive would show up as an increase in productivity in our analysis. So long as this process was slow—which seems likely—the bias in our results arising from this issue would also be small.

II.E. From When to Why

By dating the onset of productivity growth, our results help discriminate between competing explanations for why growth began. We estimate that sustained productivity growth began in England substantially before the Glorious Revolution of 1688. According to our estimates, productivity in England rose by 34% from 1600 to 1680. [North and Weingast \(1989\)](#) argue that the political regime that emerged in England after the Glorious Revolution—characterized by a power-sharing arrangement between Parliament, the Crown, and the common law courts—resulted in secure property rights and rule of law and thereby laid the foundation for economic growth. The institutional changes associated with the Glorious Revolution may well have been important for growth, but our results indicate that the seeds of growth were sown earlier.

Our results support explanations of the onset of growth that focus attention on developments that occurred in the period around 1600. The Reformation is an obvious candidate. In particular, Henry VIII's confiscation of monastic lands was a big shock to land-ownership patterns and the land market in England ([Heldring, Robinson, and Vollmer 2021](#)). England also became a favored destination for skilled immigrants fleeing religious persecution on the continent. This was also a period of rapidly increasing urbanization in England. London experienced an explosion of its population around this time—from 55,000 in 1520 to 475,000 in 1670 ([Wrigley 2010](#))—likely due to a rapid increase in international trade. English woolen exports expanded rapidly over this period (new draperies) as did intercontinental trade, colonization, and privateering. The British East India

Company was founded in 1600, and the Virginia Company founded its first permanent settlement in North America in 1607.

Our finding that the onset of growth preceded both the Glorious Revolution and the English Civil War (1642–1651) lends support to the Marxist view that economic change propelled history forward and drove political and ideological change. Marx (1867) stressed the transition from feudalism to capitalism. He argued that after the disappearance of serfdom in the fourteenth century, English peasants were expelled from their land through the enclosure movement. That spoliation inaugurated a new mode of production: one where workers did not own the means of production and could only subsist on wage labor. This proletariat was ripe for exploitation by a new class of capitalist farmers and industrialists. In that process, political revolutions were a decisive step in securing the rise of the bourgeoisie. To triumph, capitalism needed to break the remaining shackles of feudalism. As the *Communist Manifesto* puts it, “they had to be burst asunder; they were burst asunder” (Marx and Engels 1848, 40–41). Hill (1940, 1961) offers more recent treatments of the political revolutions in England in the seventeenth century that stress class conflict and their economic origins.

Acemoglu, Johnson, and Robinson (2005) synthesize the Marxist and institutionalist views. They argue that Atlantic trade enriched a merchant class that then demanded secure property rights and secured these rights through the Civil War and Glorious Revolution. This last narrative lines up well with our result that steady growth—perhaps driven by the Atlantic trade—began about half a century before the Civil War. However, we do not detect a radical increase of growth in the immediate aftermath of either the Civil War or the Glorious Revolution: 3.2% (1600–1640), 4.2% (1640–1680), and 4.5% (1680–1810).

Allen (1992) argues that a long and gradual process of institutional change in England over the 600-year period from the Norman Conquest to the Glorious Revolution resulted in a situation in the sixteenth century where the yeoman class had acquired a substantial proprietary interest in the land, and thus an incentive to innovate. The timing of Allen’s “rise of the yeoman” lines up reasonably well with our estimate of the onset of growth. According to Allen, property rights, rule of law, and personal freedom gradually expanded, and the social order was gradually transformed from a feudal to a capitalist order. From the twelfth century, royal courts helped freeholders gain full

ownership over their land. After the Black Death, serfdom collapsed as landlords competed for scarce labor. Early enclosures (fifteenth and early sixteenth centuries) involved brutal evictions and depopulation of manors. The Crown reacted to this by increasing protection of tenant farmers.

The spread of movable-type printing across Europe after 1450 led to a large increase in literacy in England in the sixteenth and seventeenth centuries (Cressy 1980; Houston 1982) and a huge drop in the price of books (Clark and Levin 2011). Dittmar (2011) argues that cities exposed to printing grew substantially faster than otherwise similar cities. More generally, Rosenberg and Birdzell (1986) argue for the emergence of institutions favorable to commerce in the early modern period in Western Europe, especially in England and Holland. These included impartial commercial courts, bills of exchange, marine insurance, and double-entry bookkeeping in addition to several of the institutions discussed above.

Printing and literacy likely had wide ranging effects on culture. Mokyr (2009, 2016) and McCloskey (2006, 2010, 2016) have argued that the crucial change that caused growth to begin was the emergence of a culture of progress based on the idea that humanity can improve its condition through science and rational thought. Others have stressed a Protestant ethic (Weber 1904, 1905) and Puritanism (Tawney 1926). The timing of these changes lines up reasonably well with our estimates, although it is not straightforward to pinpoint precisely what these theories imply about the timing of the onset of growth.

Bogart and Richardson (2011) stress the importance of the post-Glorious Revolution regime's push to reorganize and rationalize property rights through enclosures, statutory authority acts, and estate acts. While our results contradict the notion that growth began with the Glorious Revolution, the fact that England underwent massive institutional change in the seventeenth and eighteenth centuries may have played an important role in sustaining growth during this period.

Allen (2009a) argues that the Industrial Revolution occurred in Britain around 1800 because innovation was uniquely profitable then and there. His theory relies on growth in the seventeenth century leading to high real wages in England in the eighteenth century as well as the development of a large coal industry. High wages and cheap coal made it profitable to invent labor-saving technologies in textiles, such as the spinning

jenny, water frame, and mule, as well as coal-burning technologies such as the steam engine and coke smelting furnace. While our theory does not point to the Industrial Revolution as the genesis of economic growth, Allen's theory helps explain how growth was sustained and the particular direction it took that led to the huge fall in the importance of land in production that we estimate later.

III. A MALTHUSIAN MODEL OF THE ECONOMY

The simple analysis in [Section II](#) makes a number of strong assumptions. In the remainder of the article, we explore a richer framework that allows us to relax some of these. Most important, we model the evolution of the population. Thus, we can present novel estimates of the strength of Malthusian population forces in preindustrial England. We explicitly incorporate plague shocks into this part of the model, which helps the model distinguish between measurement error and true variation in the population. We also incorporate capital accumulation into the model, explore alternatives to the assumption that labor supply was proportional to the population, allow for a falling importance of land in production after the onset of industrialization, and estimate the slope of the labor demand curve using the entire data set as opposed to identifying it from the Black Death. This richer framework allows us to explore the robustness of our conclusions from [Section II](#) and present a number of additional interesting results.

While our focus is on the preindustrial period, our data extends well into the early industrial period. It is therefore important for our model to capture the character of the preindustrial economy and of the early industrial economy. The role of land in production is particularly important in this regard. Before the Industrial Revolution, land was a hugely important factor of production. The advent of steam power led to a sharp fall in the role of land in production as fossil fuels substituted for human and animal power in the production of energy (and the role of food production in the economy shrank). To capture this change, we distinguish between the pre-industrial period and the early industrial period and allow the importance of land in production to change after the onset of the Industrial Revolution. [Section III.A](#) presents our model of the preindustrial economy, and [Section III.B](#) presents our model of the early industrial economy.

III.A. A Model of the Preindustrial Economy

Output is produced with land, capital, and labor according to the production function:

$$Y_t = F_t(Z, K_t, L_t) = A_t Z^\alpha K_t^\beta L_t^{1-\alpha-\beta},$$

where K_t denotes capital.¹¹ Producer optimization gives rise to the labor demand curve:

$$W_t = (1 - \alpha - \beta) A_t Z^\alpha K_t^\beta L_t^{-\alpha-\beta}.$$

Taking logs of this equation yields

$$(3) \quad w_t = \tilde{\phi} + a_t + \beta k_t - (\alpha + \beta) l_t,$$

where $\tilde{\phi} = \log(1 - \alpha - \beta) + \alpha \log Z$.

Producers accumulate capital to the point where the marginal product of capital is equal to its user cost. This gives rise to a capital demand equation:

$$(4) \quad r_t + \delta = \beta A_t Z^\alpha K_t^{\beta-1} L_t^{1-\alpha-\beta},$$

where r_t is the rental rate for capital and δ is the rate of depreciation of capital. Because we do not have data on capital for the preindustrial period, we use the capital demand equation (equation (4)) to eliminate K_t from the labor demand equation (equation (3)). Taking logs of the resulting equation yields

$$(5) \quad w_t = \tilde{\phi}' + \frac{1}{1-\beta} a_t - \frac{\alpha}{1-\beta} l_t - \frac{\beta}{1-\beta} \log(r_t + \delta),$$

where

$$\tilde{\phi}' = \frac{\beta}{1-\beta} \log \beta + \log(1 - \alpha - \beta) + \frac{\alpha}{1-\beta} \log Z.$$

The log of productivity a_t is the sum of a permanent and transitory component:

$$(6) \quad a_t = \tilde{a}_t + \epsilon_{2t},$$

where $\epsilon_{2t} \sim \mathcal{N}(0, \sigma_{\epsilon_2}^2)$ is the transitory component and $\tilde{a}_t$ is the permanent component of productivity, which follows a random walk with drift

$$(7) \quad \tilde{a}_t = \mu + \tilde{a}_{t-1} + \epsilon_{1t},$$

with $\epsilon_{1t} \sim \mathcal{N}(0, \sigma_{\epsilon_1}^2)$. Both ϵ_{1t} and ϵ_{2t} are independently distributed over time. The transitory component of productivity may reflect

11. [Online Appendix D](#) presents results for a more general production function.

both measurement error and true transitory variation in productivity (e.g., due to transitory variation in weather). As in most macroeconomic models, productivity is a catch-all variable capturing the influence of all variables that are not explicitly modeled in the production function. Relative to the model in [Section II](#), we now incorporate capital accumulation. Our measure of productivity is narrower as a consequence. Productivity in this model nonetheless captures a number of features of reality in addition to technology, including institutions, the effects of international trade, and colonial exploitation.

The average growth rate of productivity is given by the parameter μ . As in [Section II](#), we allow for two breaks in μ , that is, two changes in the average growth rate of productivity. (We allow the variances of the transitory and permanent productivity shocks— $\sigma_{\epsilon_1}^2$ and $\sigma_{\epsilon_2}^2$ —to break at the same times.) We fix one of these breaks in 1810. This break captures the Industrial Revolution. We allow for another break earlier in the sample between 1550 and 1800.

To pin down the timing of the first break, we estimate a mixture model. Since μ , $\sigma_{\epsilon_1}^2$, and $\sigma_{\epsilon_2}^2$ break twice, they take on three values, one for each regime. We denote these as $\mu(i)$ with $i \in \{1, 2, 3\}$ (with analogous notation for $\sigma_{\epsilon_1}^2$ and $\sigma_{\epsilon_2}^2$). From the beginning of our sample until 1540, $\mu = \mu(1)$. From 1550 until 1800, $\mu = (1 - I)\mu(1) + I\mu(2)$, where I is an indicator variable that switches from zero to one at the time of the first break. Finally, from 1810 until 1860, $\mu = \mu(3)$. The indicator variable I has a multinomial distribution with probabilities of switching from zero to one at each date between 1550 and 1800. We estimate the probabilities of the multinomial distribution for I . The prior for these probabilities is a Dirichlet distribution with concentration vector $c_b \times (1, \dots, 1)$. We choose a small value for c_b . This ensures that each draw from the distribution is close to a corner of the distribution, that is, chooses a specific break date. In particular, we set $c_b = 0.001$.¹² The output from our estimation of these probabilities is a posterior probability distribution over break dates.

We assume that the labor force in the economy is proportional to the population and that each worker works D_t days per year.

12. For a simple exposition of the Dirichlet distribution and the role of the concentration parameter, see Stan Function Reference, section 23: <https://mc-stan.org/docs/functions-reference/dirichlet-distribution.html>.

This implies that

$$L_t = \Lambda D_t N_t,$$

where N_t denotes the population and Λ is a constant. Taking logs of this equation and using the resulting equation to eliminate l_t in [equation \(5\)](#) yields

$$(8) \quad w_t = \phi + \frac{1}{1-\beta} a_t - \frac{\alpha}{1-\beta} (d_t + n_t) - \frac{\beta}{1-\beta} \log(r_t + \delta),$$

where $\phi = \tilde{\phi}' - \frac{\alpha}{1-\beta} \lambda$ with $\lambda = \log \Lambda$.

A central aspect of our model is the law of motion for the population. Following [Malthus \(1798\)](#), we assume that population growth is increasing in real income:

$$\frac{N_t}{N_{t-1}} = \Omega (W_{t-1} D_{t-1})^\gamma \Xi_t,$$

where Ω is a constant, γ is the elasticity of (gross) population growth with respect to real income, and Ξ_t denotes other (exogenous) factors affecting population growth. Taking logs of this equation yields

$$(9) \quad n_t - n_{t-1} = \omega + \gamma(w_{t-1} + d_{t-1}) + \xi_t.$$

Malthus argued that the birth rate and the death rate varied with real income. He described “preventive checks” on population growth that lowered birth rates. These included contraception, delayed marriage, and regulation of sexual activity during marriage. Malthus also described “positive checks” on population growth that raised death rates. These include disease, war, severe labor, and extreme poverty. Modern research on fertility emphasizes a quantity-quality trade-off where parents invest in child quality rather than child quantity when their income rises ([Becker and Lewis 1973](#); [Becker and Barro 1988](#)). In our model, the parameter γ captures the combined effect of all these effects.

We allow for two types of exogenous population shocks:

$$(10) \quad \xi_t = \xi_{1t} + \xi_{2t}.$$

First, we allow for “plague” shocks:

$$(11) \quad \exp(\xi_{1t}) \sim \begin{cases} \beta(\beta_1, \beta_2), & \text{with probability } \pi \\ 1, & \text{with probability } 1 - \pi. \end{cases}$$

These plague shocks occur infrequently (with probability π) but when they occur, they kill a (potentially sizable) fraction of the population. The fraction of the population that survives follows

TABLE II
ENERGY CONSUMPTION (PETAJOULES) IN ENGLAND AND WALES

	1560	1700	1750	1800	1850
Draught animals	21.1	32.8	33.6	34.3	50.1
People	14.9	27.3	29.7	41.8	67.8
Firewood	21.5	22.5	22.6	18.5	2.2
Wind	0.2	1.4	2.8	12.7	24.4
Water	0.6	1.0	1.3	1.1	1.7
Coal	6.9	84.0	140.8	408.7	1,689.1
Total	65.1	168.9	230.9	517.1	1,835.3
Total less coal	58.2	84.9	90.1	108.4	146.2

Notes. The table shows energy consumption in England and Wales from various sources in petajoules. It reproduces a portion of Table 2.1 in [Wrigley \(2010\)](#).

a beta distribution $\beta(\beta_1, \beta_2)$. We also allow for a second type of population shock: $\xi_{2t} \sim \mathcal{N}(0, \sigma_{\xi_2}^2)$. Both population shocks are independently distributed over time. Together, these population shocks are meant to capture a host of potential influences on population growth, in addition to plagues.

III.B. A Model of the Early Industrial Economy

The last century of our data covers the early industrial period in England. A crucial development over this time was the rapid fall in the importance of land as a factor of production. The primary driving force in this development was the introduction of the steam engine, powered by fossil fuels. This technological advance meant that the production of energy was no longer land intensive ([Wrigley 2010](#)). [Table II](#) presents data on energy consumption from various sources over our sample period. Prior to the introduction of steam power, the vast majority of energy was derived from draught animals, human power, and firewood. These energy sources were extremely land intensive. Both draught animals and human power rely on the production of food (feed in the case of animals), and firewood requires vast forests. In this environment, the fixed nature of land created a severe bottleneck for economic expansion. Our preindustrial model captures this by explicitly modeling the reliance of production on land.

With the introduction of the steam engine, energy production was gradually decoupled from land use. This dramatically reduced the importance of land in production. The last two columns in [Table II](#) show just how enormous the increase in energy consumption was during even the relatively early phase of the

industrial era. By 1850, total energy consumption had risen by over a factor of 10 relative to 1700, with virtually all of this increase coming from coal. It is clear that nothing remotely like this would have been possible without steam power or some other dramatically less land-intensive energy source.

To capture this change parsimoniously, we allow the exponents in the production function to change after the onset of the Industrial Revolution. [Acemoglu and Restrepo \(2018, 2019b\)](#) show how a production function that is explicit about how intermediate inputs (such as energy) produced by completing various tasks and new technologies (such as the steam engine) can change the factor content of production is equivalent to a traditional production function written in terms of primary factors only as long as the exponents on the factors are allowed to vary with technical progress. We take this approach and assume for the early industrial period that

$$(12) \quad Y_t = F_t(Z, K_t, L_t) = A_t Z^{\alpha_t} K_t^{\beta_t} L_t^{1-\alpha_t-\beta_t},$$

which is the same as before except that the exponents α_t and β_t are now time varying. A fall in α_t will then capture the fall in the importance of land as a factor of production, and the evolution of β_t will determine to what extent it is capital or labor that increases in importance. We do not model the use of fossil fuels explicitly. Their use is reflected in the changing output elasticities and in higher productivity.

In this case, the labor demand curve ([equation \(8\)](#)) becomes:

$$(13) \quad w_t = \phi_t + \frac{1}{1-\beta_t} \alpha_t - \frac{\alpha_t}{1-\beta_t} (d_t + n_t) - \frac{\beta_t}{1-\beta_t} \log(r_t + \delta),$$

where

$$\phi_t = \frac{\beta_t}{1-\beta_t} \log \beta_t + \log(1-\alpha_t-\beta_t) + \frac{\alpha_t}{1-\beta_t} \log Z - (\alpha_t + \beta_t) \lambda.$$

The fact that α_t and β_t can change implies that more data is needed to identify the model: an increase in wages for a given level of labor supply could be due to an increase in α_t or a fall in α_t . To address this issue, we make use of the demand curves for land and capital and data on the quantity of capital and the price of land after 1760. We assume that potential renters and owners of land will trade until the rental price of land equals its marginal product:

$$(14) \quad S_t = \alpha_t A_t Z^{\alpha_t-1} K_t^{\beta_t} L_t^{1-\alpha_t-\beta_t},$$

where S_t denotes the rental price of land. Capital demand is the same as before (equation (4)) except that α_t and β_t are time varying. Taking logs and manipulating the land, capital, and labor demand curves yields:

$$(15) \quad s_t = w_t + n_t + d_t - \log Z + \log \alpha_t - \log(1 - \alpha_t - \beta_t) + \lambda$$

$$(16) \quad \begin{aligned} \log(r_t + \delta) &= w_t + n_t + d_t - k_t + \log \beta_t \\ &\quad - \log(1 - \alpha_t - \beta_t) + \lambda. \end{aligned}$$

These two extra equations pin down α_t and β_t . Simple manipulation of the demand curves for labor, land, and capital—which we spell out in detail in [Online Appendix E](#)—establishes three intuitive results. First, an increase in s_t (land rents) holding other variables constant implies an increase in α_t and a decrease in β_t and $1 - \alpha_t - \beta_t$. Second, an increase in r_t (the return on capital) holding other variables constant implies an increase in β_t and a decrease in α_t and $1 - \alpha_t - \beta_t$. Third, an increase in w_t (wages) holding other variables constant implies an increase in $1 - \alpha_t - \beta_t$ and a decrease in α_t and β_t . At a mechanical level, we are adding two equations per time period—equations (15) and (16)—and two observable variables per time period— s_t and k_t .

III.C. Measuring Productivity with Structural Transformation

Allowing the exponents in the production function to change raises an important complication. In this case, A_t is no longer a natural measure of productivity.¹³ Productivity is meant to capture the rate at which inputs can be converted into outputs. In settings with multiple inputs (or outputs), how to operationalize this concept is ambiguous. In some cases, such as $Y_t = A_t F(X_t)$ where X_t denotes a vector of inputs, all reasonable measures of productivity agree (in this case A_t). But in the more general case of $Y_t = F_t(X_t)$, this is not the case. [Caves, Christensen, and Diewert \(1982\)](#) introduce the notion of a Malmquist productivity index for a quite general case of production technologies, based on ideas in

13. One simple way to see this is to consider a change in the units that we use to express labor. Suppose $\tilde{L}_t \equiv \psi L_t$, then $Y_t = A_t Z^{\alpha_t} K_t^{\beta_t} \tilde{L}_t^{1-\alpha_t-\beta_t} = \frac{A_t}{\psi^{1-\alpha_t-\beta_t}} Z^{\alpha_t} K_t^{\beta_t} \tilde{L}_t^{1-\alpha_t-\beta_t}$. With the new units for labor, it is $\frac{A_t}{\psi^{1-\alpha_t-\beta_t}}$ rather than A_t that multiplies the factors of production. If α_t and β_t change over time, $\frac{A_t}{\psi^{1-\alpha_t-\beta_t}}$ will behave differently from A_t . Clearly, a more general concept of productivity is needed. See [Online Appendix F](#) for a discussion of how the Malmquist index (introduced below) avoids this issue.

Malmquist (1953). This index uses the notion of the distance between the input-output vector chosen at one point in time and the technological frontier at another point in time. For example, the distance of the input-output vector that is chosen in period $t + 1$ from the time t technological frontier is $D_t(X_{t+1}, Y_{t+1}) = \frac{F_{t+1}(X_{t+1})}{F_t(X_{t+1})}$ — that is, actual output at time $t + 1$ divided by counterfactual output using the input vector of time $t + 1$ but the time t production function. Analogously, the distance of the input-output vector that is chosen at time t from the time $t + 1$ technological frontier is $D_{t+1}(X_t, Y_t) = \frac{F_t(X_t)}{F_{t+1}(X_t)}$.

Caves, Christensen, and Diewert (1982) recommend using a geometric average of $D_t(X_{t+1}, Y_{t+1})$ and $D_{t+1}(X_t, Y_t)^{-1}$ as a Malmquist index of productivity:

$$(17) \quad \frac{M_t}{M_{t-1}} = \sqrt{\frac{F_t(Z, K_t, L_t)F_t(Z, K_{t-1}, L_{t-1})}{F_{t-1}(Z, K_t, L_t)F_{t-1}(Z, K_{t-1}, L_{t-1})}}, \quad M_0 = 1.$$

We adopt this recommendation. With constant exponents α and β , the growth rates of M_t and A_t are the same (i.e., $\frac{M_t}{M_{t-1}} = \frac{A_t}{A_{t-1}}$). More generally, the growth rate of M_t will differ from the growth rate of A_t in important ways. See [Online Appendix F](#) for a more detailed discussion of the Malmquist index.

With production function (12), the log of the Malmquist index is

$$(18) \quad \hat{m}_t = \hat{\alpha}_t + \hat{\alpha}_t \log Z + \hat{\beta}_t \bar{k}_t - (\hat{\alpha}_t + \hat{\beta}_t)(\bar{d}_t + \bar{n}_t + \lambda),$$

where hats denote deviations from the previous period, $\hat{x}_t = x_t - x_{t-1}$, and bars denote the average of period $t - 1$ and period t , $\bar{x}_t = \frac{x_{t-1} + x_t}{2}$. Once again, if α_t and β_t are constant, this expression collapses to $\hat{m}_t = \hat{\alpha}_t$.

As in the preindustrial era, we assume that the log of productivity (m_t) is subject to permanent and transitory shocks in the early industrial era:

$$(19) \quad m_t = \tilde{m}_t + \epsilon_{2t},$$

where

$$(20) \quad \tilde{m}_t = \mu + \tilde{m}_{t-1} + \epsilon_{1t},$$

$\epsilon_{1t} \sim \mathcal{N}(0, \sigma_{\epsilon_1}^2)$, $\epsilon_{2t} \sim \mathcal{N}(0, \sigma_{\epsilon_2}^2)$, and ϵ_{1t} and ϵ_{2t} are independently distributed over time. Here, $\tilde{m}_t$ is the permanent component of productivity, which follows a random walk with drift, while ϵ_{2t} is the transitory component of productivity.

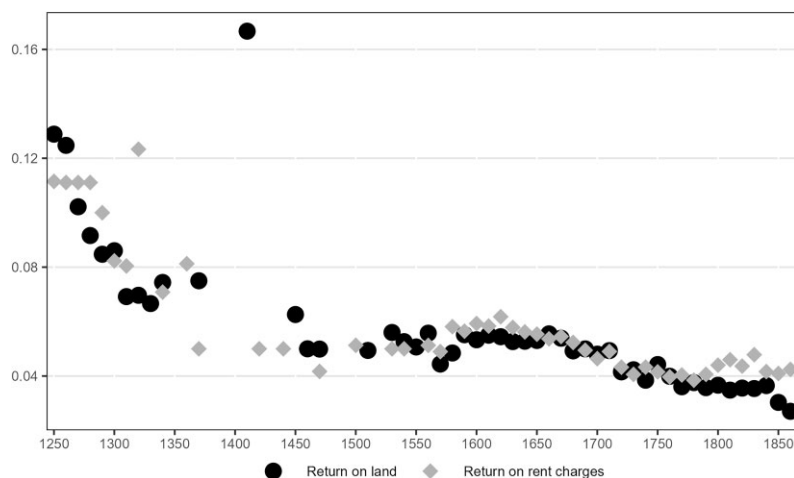


FIGURE VIII

Rates of Return on Land and Rent Charges

The figure plots the data we use on rates of return on land and rent charges. These data are from [Clark \(2002, 2010\)](#).

III.D. Data on Interest Rates, Land Rents, Capital, and Days Worked

For the preindustrial period, we use data on rates of return to estimate the evolution of the capital stock. [Figure VIII](#) plots our data on rates of return on agricultural land and “rent charges” compiled by [Clark \(2002, 2010\)](#). The rate of return on agricultural land is measured as $\frac{R}{P}$, where R is the rent and P is the price of a piece of land. “Rent charges” should not be confused with land rents. Rent charges were yields on perpetual nominal debt obligations secured by land or buildings (i.e., a collateralized loan). These are also measured as $\frac{R}{P}$, where R is the annual payment and P is the price of the obligation (which was usually much smaller than the value of the collateral). See [Clark \(2010\)](#) for more detail.

We view our series on rates of return of agricultural land and rent charges as two noisy measures of the rate of return on capital in England over our sample period. In other words, we assume that $r_t = \tilde{r}_{it} + \iota_{it}^r$, where r_t denotes the true rate of return on capital at time t , $\tilde{r}_{it}$ denotes noisy measure i , and $\iota_{it}^r \sim t_{vir}(0, \tilde{\sigma}_{ir}^2)$ denotes measurement error. In periods when neither measure is

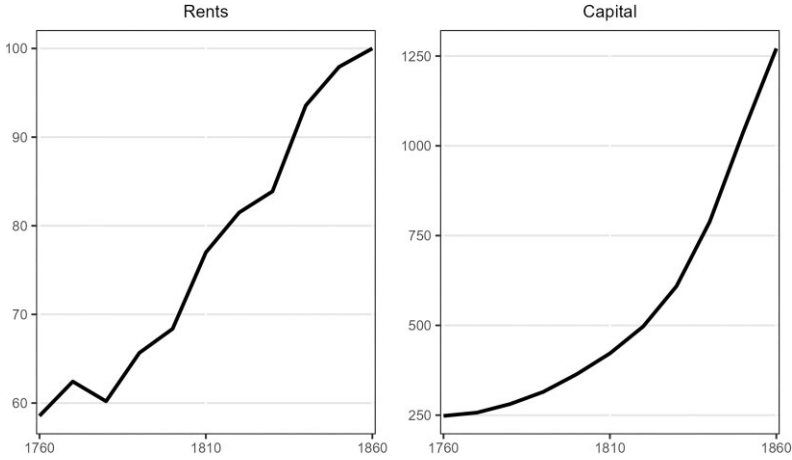


FIGURE IX

Land Rents and Capital after 1760

The figure plots the data we use on land rents from [Clark \(2002, 2010\)](#) and the net capital stock from [Feinstein \(1988, Table VIII\)](#).

available, we assume that the interest rate follows a random walk with truncated normal innovations: $r_t \sim \mathcal{N}_{(0,2)}(r_{t-1}, 0.01^2)$.

We date the shift from the preindustrial period to the early industrial period to between 1760 and 1770. This is a traditional date for the beginning of the Industrial Revolution. Also, estimating a changing production function—changes in α_t and β_t —requires systematic data on rents and the capital stock. The capital stock series we use is from [Feinstein \(1988\)](#) and is only available after 1760. [Figure IX](#) plots the data on land rents and the capital stock that we use. The land rents series we use is an index from [Clark \(2002, 2010\)](#). Feinstein's series is for the net capital stock and is expressed in millions of pounds in 1851–1860 prices. It reflects both industrial and agricultural investment. We assume that both variables are observed with measurement error $s_t = \tilde{s}_t + \iota_t^s$ and $k_t = \tilde{k}_t + \iota_t^k$, where s_t and k_t denote the true land rent and capital stock, respectively; $\tilde{s}_t$ and $\tilde{k}_t$ denote our noisy measures of land rents and the capital stock, respectively; and $\iota_t^s \sim t_{\nu_s}(0, \tilde{\sigma}_s^2)$ and $\iota_t^k \sim t_{\nu_k}(0, \tilde{\sigma}_k^2)$ denote the measurement error in these variables.

TABLE III
PRIORS FOR MODEL PARAMETERS

Parameter	Prior	Parameter	Prior
β	$\mathcal{U}(0, 1)$	γ	$\mathcal{U}(-2, 2)$
φ^x	$\mathcal{N}(0, 100^2)$	ψ	$\mathcal{N}(10.86, 0.07^2)$
ω	$\mathcal{N}(0, 1)$	μ	$\mathcal{N}(0, 1)$
μ_{ξ_1}	$\mathcal{U}(0.5, 0.9)$	v_{ξ_1}	$\mathcal{P}_I(1, 1.5)$
π	$\mathcal{U}(0, 0.5)$	δ	$\mathcal{N}_{(0,0.2)}(0.1, 0.05^2)$
$\sigma_{\varepsilon_1}^2$	$\Gamma(3, 0.001)$	$\sigma_{\varepsilon_2}^2$	$\Gamma(3, 0.005)$
$\sigma_{\varepsilon_2}^2$	$\Gamma(3, 0.005)$	σ_n^2	$\Gamma(3, 0.005)$
$\sigma_{\varepsilon_3}^2$	$\Gamma(3, 0.005)$	σ_{rj}^2	$\Gamma(3, 0.005)$
$\tilde{\sigma}_2^2$	$\Gamma(3, 0.005)$	$\tilde{\sigma}_s^2$	$\Gamma(3, 0.005)$
$\tilde{\sigma}_k^2$	$\Gamma(3, 0.005)$	v_n^{-1}	$\mathcal{U}(0, 1)$
v_d^{-1}	$\mathcal{U}(0, 1)$	v_s^{-1}	$\mathcal{U}(0, 1)$
v_k^{-1}	$\mathcal{U}(0, 1)$	v_{ir}^{-1}	$\mathcal{U}(0, 1)$

III.E. Priors and Estimation Details

We use Bayesian methods to estimate our model. In particular, we use a Hamiltonian Monte Carlo sampling procedure (Gelman et al. 2013; Betancourt 2018).¹⁴ Table III lists the priors we assume for the model parameters. In all cases, we choose highly dispersed priors. Most of the priors are self-explanatory, but some comments are in order. Our baseline estimation for the slope of the labor demand curve is based on movements in real wages and the population between 1340 and 1360 as discussed in Section II. With capital, the slope of the labor demand curve becomes $\tilde{\alpha} = \frac{\alpha}{1-\beta}$. We set a uniform prior for β between zero and one and recover α from $\alpha = \tilde{\alpha}(1 - \beta)$. The prior listed for β in Table III is for this case. We also present results where (α, β) are estimated structurally from our full model. We explain the prior for that case below.

The prior for ψ is set such that the peak population before the Black Death is between 4.5 and 6 million with 95% probability. This range encompasses the estimates of Clark (2007b) and Broadberry et al. (2015). Rather than specifying priors for β_1 and β_2 , we specify priors for the mean of ξ_1 which we denote $\mu_{\xi_1} = \frac{\beta_1}{\beta_1 + \beta_2}$ and the pseudo sample size of ξ_1 which we denote $v_{\xi_1} = \beta_1 + \beta_2$. The priors we choose for these parameters follow the recommendations of Gelman et al. (2013, 110) for a flat prior

14. We implement this procedure using Stan (Stan Development Team 2017).

for a beta distribution. [Online Appendix](#) Figure A.3 plots the prior densities for the standard deviations of ϵ_1 , ϵ_2 , and ξ_2 . In [Section IV.C](#), we discuss how varying our priors affects our main results.

To discipline the behavior of α_t and β_t after 1760, we assume that the simplex $(\alpha_t, \beta_t, 1 - \alpha_t - \beta_t)$ follows a Dirichlet distribution with concentration vector $c_s \times (\alpha_{t-1}, \beta_{t-1}, 1 - \alpha_{t-1} - \beta_{t-1})$, where $c_s = 3$. For any value of c_s , this implies that the mean of α_t is α_{t-1} and the mean of β_t is β_{t-1} . The choice $c_s = 3$ implies that, with $\alpha_{t-1} = \beta_{t-1} = \frac{1}{3}$, the distribution is uniform over simplexes. A smaller value of c_s would concentrate the prior distribution toward the corners of the simplex—draws where one of the coefficients is close to one and the others close to zero. A larger value, on the other hand, concentrates the prior toward the mean of the distribution—most draws would be close to $(\alpha_{t-1}, \beta_{t-1}, 1 - \alpha_{t-1} - \beta_{t-1})$. Thus, our prior choice is a way to center the distribution around the previous value of the coefficients, while allowing them to change if the likelihood dictates it. In the case where (α, β) is estimated structurally before 1760, our prior distribution for $(\alpha, \beta, 1 - \alpha - \beta)$ is a Dirichlet distribution with concentration vector $(1, 1, 1) = 3 \times (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ which, once again, corresponds to a uniform distribution over simplexes.

We allow for a structural break in the probability of a plague π in 1680. The timing of this break is chosen to immediately follow the Great London Plague of 1665.¹⁵ This break is meant to capture the fact that plagues are less frequent in the latter part of our sample. The exact timing of this break does not affect our main results in a material way.

Finally, we need to normalize some of the variables in our model. Labor demand ([equation \(13\)](#)) features $\log Z$ and λ , which we normalize to zero. The observed wage, land rents, and our capital series are indices. They are thus not normalized in a theoretically consistent fashion. For wages, for instance, we would want to observe the marginal product of an additional unit of labor expressed in units of aggregate output. What we observe is an index that is normalized to 100 in 1860. Our Bayesian framework allows us to handle this issue in a straightforward fashion. For wages we assume that $w_t = \varphi^w + \tilde{w}_t$, where $\tilde{w}_t$ is the observed

15. Notice that the change in the population between the 1660s and the 1670s is affected by the Great London Plague. So ξ_{1t} for $t = 1670$ will be affected by the Great London Plague. This is why we assume that ξ_{1t} for $t \geq 1680$ is governed by a different π than earlier values of ξ_{1t} .

wage, w_t is the true wage, and $\varphi^w \sim \mathcal{N}(0, 100^2)$ is a normalization constant. For land rents and the capital stock, we add the following normalization constants to their measurement equations: $\varphi^s \sim \mathcal{N}(0, 100^2)$ and $\varphi^k \sim \mathcal{N}(0, 100^2)$, respectively. We reproduce the equations and distributional assumptions of our full model in [Online Appendix G](#) for convenience.

IV. ESTIMATES OF PRODUCTIVITY FROM THE FULL MALTHUSIAN MODEL

We present results on the evolution of productivity in England for several variants of the model presented in [Section III](#). We consider a case in which we maintain our earlier assumptions that the importance of land in production (α) is constant, days worked are constant, and that the slope of the labor demand curve is pinned down by movements in wages and the population at the time of the Black Death. This case differs from the simple model presented in [Section II](#) in that we have added capital and the Malthusian model of population dynamics. We refer to this case as the constant α , β case.

[Figure X](#) presents results on the evolution of productivity over time for the constant α , β case (gray solid line). We also include results for the simple model from [Section II](#) for comparison (black solid line). While the overall pattern is similar, the constant α , β case of the full model implies slower productivity growth than the simple model. The reason for this is that the full model incorporates capital accumulation. As a result, a portion of the increase in labor demand over our sample period is attributed to capital accumulation as opposed to productivity and less is left in the residual to be attributed to productivity growth.¹⁶

IV.A. *Productivity and the Falling Importance of Land*

The models we have discussed so far make the standard but highly unrealistic assumption that the elasticity of output with respect to land α is constant through the early industrial period up to 1870. We relax this assumption by allowing the production function parameters α and β to change over time after the onset

16. [Online Appendix](#) Figure A.2 presents our estimate of the timing of the preindustrial structural break in productivity growth in the constant α , β case, our baseline case discussed below, and several other variants of the model. All these cases strongly favor a break in 1600.

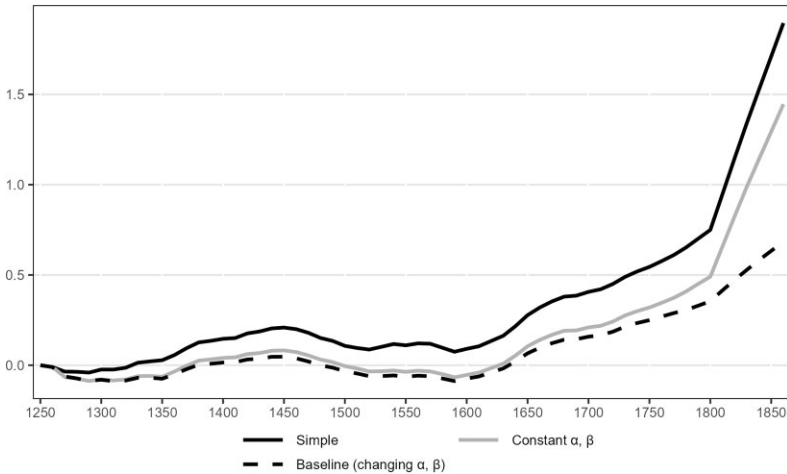


FIGURE X
Permanent Component of Productivity

The figure plots our estimates of the evolution of the permanent component of productivity $\hat{a}_t$ over the sample period for several variants of our full Malthusian model (natural log relative to its value in 1250).

of the Industrial Revolution. As described in [Section III](#), we use data on the quantity of capital and the price of land after 1760 to pin down the evolution of α_t and β_t during the early industrial period.

[Figure XI](#) plots our estimates of the evolution of α_t and β_t over time. Recall that α_t corresponds to the elasticity of output with respect to land, and β_t corresponds to the elasticity of output with respect to capital. We estimate a value for α before the Industrial Revolution of 0.54 (with a standard deviation of 0.05) and a value for β of 0.23 (with a standard deviation of 0.07).¹⁷ After 1760, we estimate a sharp fall in α_t . By 1860, α_t had fallen by roughly half to a value of 0.29. This sharp fall in α_t reflects the nature of technical change during the Industrial Revolution. As we discuss in [Section III](#), the advent of the steam engine powered

17. As we discuss in [Section II](#), the land share of production differs from α when the elasticity of substitution between land and labor differs from one. In particular, if land and labor are complements in production, the land share is lower than α . Explicitly, allowing for a production function that is more general than the Cobb-Douglas production function is complicated when the structure of the production function is changing as it is for us after 1760.

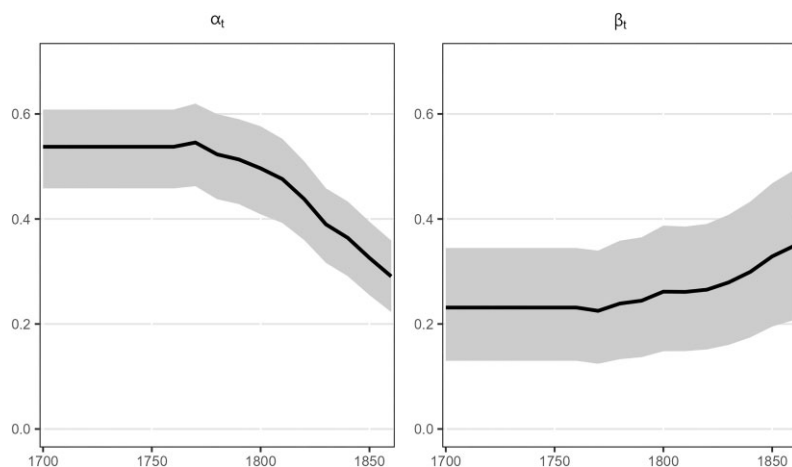


FIGURE XI

Parameters of the Production Function

The figure plots our estimates of the evolution of the parameters α_t and β_t in our baseline case. They are constant until 1770 by assumption. The black line is the mean of the posterior for each period, and the gray shaded area is the 90% central posterior interval.

by fossil fuels meant that the production of energy was no longer land intensive. This led to a large fall in the importance of land in production. As α_t falls, both β_t and $1 - \alpha_t - \beta_t$ (the elasticity of output with respect to labor) rise. Between 1760 and 1860, β_t increased from 0.23 to 0.35, and $1 - \alpha_t - \beta_t$ increased from 0.23 to 0.36.¹⁸

Allowing for a fall in the importance of land in production after 1760 has a substantial effect on our estimate of productivity growth after that date. The dashed black line in [Figure X](#) plots our estimates of productivity growth in this case, which we refer to as our baseline case. Productivity growth up until 1760 is very similar to the constant α , β case, but after 1760 it is much slower. In the constant α , β case, our estimate of average productivity growth between 1800 and 1870 is 16% per decade ([Table IV](#)). When we allow α_t and β_t to change in the early industrial period, our estimate of average productivity growth falls to a more modest level of 5% per decade over this period. This baseline case

18. Estimates of μ and γ are discussed below. Estimates of other model parameters are presented in [Online Appendix Table A.1](#).

TABLE IV
PRODUCTIVITY GROWTH

	Mean	Std. dev.	2.5%	97.5%
		Simple		
$\mu_{a,1}$	0.00	0.01	-0.01	0.02
$\mu_{a,2}$	0.04	0.02	0.02	0.10
$\mu_{a,3}$	0.19	0.01	0.17	0.22
		Constant α, β		
$\mu_{a,1}$	-0.00	0.01	-0.01	0.01
$\mu_{a,2}$	0.03	0.01	0.01	0.05
$\mu_{a,3}$	0.16	0.02	0.11	0.20
		Baseline		
$\mu_{a,1}$	-0.00	0.01	-0.01	0.01
$\mu_{a,2}$	0.02	0.01	0.01	0.04
$\mu_{a,3}$	0.05	0.01	0.03	0.08

Notes. The table presents the mean, standard deviation, 2.5% quantile, and 97.5% quantile of the posterior distribution we estimate for average productivity growth μ for three cases of our model. See [Online Appendix Table A.1](#) for estimates of the posterior distribution of other model parameters.

is the specification used in [Figures I and II](#) of the introduction. Also, the numerical estimates discussed in the abstract and introduction are for this case.

Intuitively, in the constant α, β case with a large α , the marginal product of labor is sharply downward sloping. After 1760, England experienced explosive growth in its population ([Figure IV](#)). This led to strong downward pressure on real wages. Since real wages actually rose over this period, large increases in productivity are needed to fit the data. When α is allowed to fall after 1760, the labor demand curve becomes less downward sloping, which implies that the downward pressure on real wages from population growth is smaller, and therefore less productivity growth is needed to explain the increase in real wages.

The model with falling importance of land after 1760 attributes a substantial portion of the explosive economic growth of the early industrial period not to productivity growth but to structural change. Before the Industrial Revolution, land was a severe bottleneck for economic growth. The advent of the coal-powered steam engine changed this dramatically, freeing the economy to grow more rapidly. Whether one wants to view this change as an increase in productivity (as we do in the constant α, β case) or as structural change is a matter of how detailed a model one considers. Productivity captures both forces in the simple model. In the more detailed model here, we provide an explanation for a large



FIGURE XII

Days Worked per Worker in England, 1260–1840

The figure presents an estimate of the evolution of days worked per worker in England from [Humphries and Weisdorf \(2019\)](#).

chunk of this variation in the early industrial period, through the lens of structural change.

The data we use on the capital stock and land rents starts in 1760. We view these variables as being unobserved prior to that time. [Online Appendix](#) Figure A.4 plots what our model implies about their evolution over our full sample period. We infer that capital grew at a modest pace before 1600, with faster growth thereafter. We infer relatively stable land rents prior to 1600 with growth thereafter.

IV.B. *Days Worked and Structural Identification of the Slope of Labor Demand*

The results we have presented up to this point have made the assumption that labor supply was proportional to the population of England over our sample. This is a common assumption in the literature. However, recent work by [de Vries \(2008\)](#) and [Humphries and Weisdorf \(2019\)](#) argues that days worked per worker fluctuated substantially over our sample. [Figure XII](#) plots Humphries and Weisdorf's series. It indicates that days worked dropped sharply after the Black Death and then recovered to its

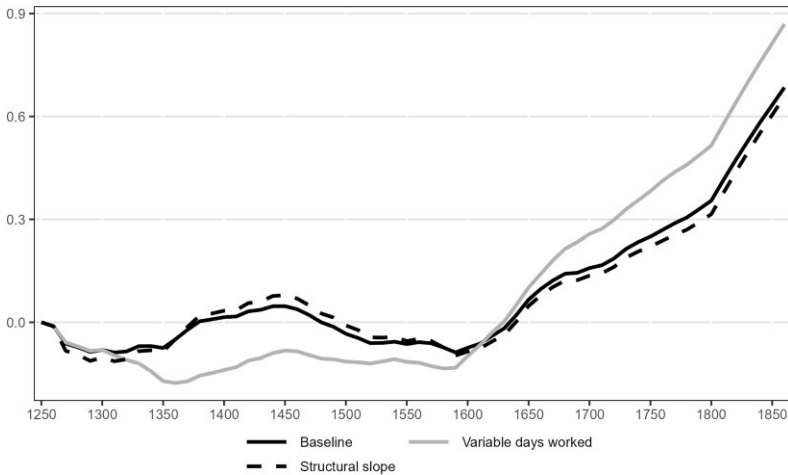


FIGURE XIII

Permanent Component of Productivity

The figure plots our estimates of the evolution of the permanent component of productivity $\hat{a}_t$ over our sample period for several variants of our full Malthusian model (natural log relative to its value in 1250).

previous level by the early seventeenth century. After that, days worked kept increasing, rising far above their previous level.¹⁹

Figure XIII presents results on productivity for a case where we incorporate Humphries and Weisdorf's (2019) estimates of days worked into our analysis. Here, we assume that days worked are exogenous and are measured with error $d_t = \tilde{d}_t + \epsilon_t^d$, where d_t denotes the true number of days worked per worker, which are unobserved; $\tilde{d}_t$ denotes Humphries and Weisdorf's estimates of

19. Other researchers have referred to the increase in worker industriousness over this period as an Industrious Revolution (de Vries 1994, 2008). However, the degree to which days worked changed over time in England is controversial. Comparisons of direct estimates by Blanchard (1978) for 1400–1600 and Voth (2000, 2001) for 1760–1830 support the idea that days worked were low in the post-Black Death period and rose sharply in the seventeenth and eighteenth centuries. Earlier indirect estimates by Clark and Van Der Werf (1998), however, suggest modest changes in days worked over our sample. Humphries and Weisdorf (2019) argue that their new series on the income of workers on annual contracts represents an important improvement relative to the series used by Clark and Van Der Werf (1998).

days worked; and $\epsilon_t^d \sim t_{v_d}(0, \hat{\sigma}_d^2)$ denotes the measurement error.²⁰ Allowing for changing days worked does not affect our conclusion about the timing of the onset of productivity growth. However, we estimate a somewhat larger increase in average productivity growth after 1600 in this case. Intuitively, more productivity is needed to compensate for the additional labor supply associated with increased days of work. With Humphries and Weisdorf's estimates of variable work hours, we estimate that average productivity growth μ is 3% per decade for 1600–1800 and 6% per decade for 1810–1870.

We have used the Black Death as an exogenous shock to identify the slope of the labor demand curve in the preindustrial era. We also consider the alternative approach of using the structure of the full model to identify this slope. In this case, we specify a relatively diffuse prior for both α and β and use Bayesian updating to calculate a posterior mean for these and for the slope of the labor demand curve conditional on the entire sample, as we do for other parameters. We are then relying on the Malthusian model of population dynamics to account for the endogeneity of the population. [Figure XIII](#) presents results on productivity for this case. The results are very similar to the baseline case.

IV.C. Robustness

1. *Alternative Real Wage Data.* Our results up to this point use real wage data for unskilled builders from [Clark \(2010\)](#). [Online Appendix](#) Figure A.5 presents five alternative productivity series where we instead use other wage series. First, we present estimates of productivity using the following day wage series: (i) [Clark's \(2010\)](#) real wages series for farm laborers, (ii) [Clark's \(2010\)](#) real wages series for building craftsmen, and (iii) [Allen's \(2007\)](#) real wage series for the period 1770 onward (with our baseline wage series before that time). We present estimates of productivity based on the assumption that the builders, farmers, and craftsmen series from [Clark \(2010\)](#) are all noisy signals of the underlying true wage. Finally, we use [Humphries and Weisdorf's \(2019\)](#) annual income series and the assumption that days worked are constant. We do this robustness analysis for the structural-slope case. These alternative productivity series yield

20. Humphries and Weisdorf do not provide estimates for 1250, 1850, and 1860. We extrapolate days worked on these dates assuming that $d_t = d_{t-1} + \eta_t$ where $\eta_t \sim \mathcal{N}(0, \sigma_d^2)$.

similar results to the baseline productivity series, although there is some divergence early in the sample period.

2. *Population Data.* Our results up to this point use population data prior to 1540 from [Clark \(2010\)](#). [Online Appendix Figure A.6](#) presents estimates of productivity using population data from [Broadberry et al. \(2015\)](#) for the period before 1540. [Broadberry et al.'s \(2015\)](#) estimates of the population are infrequent and irregular in their frequency. There are quite a few decades for which [Broadberry et al. \(2015\)](#) have no estimate, for example, they present no estimate between 1450 and 1522. In this robustness analysis, we view the population as an unobserved variable in decades for which we do not have an estimate from [Broadberry et al. \(2015\)](#). Our results on the evolution of productivity for this case are very similar to our baseline model.

3. *Break Dates.* [Online Appendix Figure A.7](#) compares the evolution of the permanent component of productivity for four different assumptions about when the pre-Industrial Revolution break to the productivity component occurred. Recall that [Figure X](#) presents the evolution of the permanent component of productivity integrating over the probability of breaks occurring at different dates as in [Online Appendix Figure A.2](#). [Online Appendix Figure A.7](#) compares these results with results when we condition on the break occurring at a specific date. We do this for the baseline case and present results for several different dates between 1550 and 1760. These alternative results are very similar.

4. *Priors.* [Online Appendix Figure A.8](#) presents estimates of productivity using different prior distributions than we use in our main analysis. First, we present results for a case where we change the prior on σ_{ϵ_1} —the variance of permanent productivity shocks—to be $\text{IG}(3, 0.005)$, that is, the same as the prior on the other productivity and population shocks. Second, we present results for a case where we change the prior on ψ —the level of the population before 1540—to be $\mathcal{N}(10.86, 10^2)$, that is, much wider than in our main analysis. In both cases, the resulting productivity series are very similar to our main results. Other priors are quite dispersed.

5. *Comparison with Clark (2010, 2016).* Our estimates of productivity differ substantially from those of Clark (2010, 2016). Online Appendix H presents a detailed decomposition of the factors leading to the differences. Several factors are important. One important contributor to the difference between our series and Clark's (better known) 2010 series is an error in that series that leads to a spurious 25 log point drop between 1540 and 1550. Using the average of factor output elasticities at time t and $t - 1$ when calculating changes in productivity between time t and $t - 1$ also explains an important part of the difference in our results relative to Clark's, especially for the period before 1600. Differences in the factor prices and factor output elasticities implied by our approach, relative to those used by Clark, explain the remaining differences.

V. LIBERATING THE ECONOMY FROM THE IRON LAW OF WAGES

In Section IV, we estimate a gradual increase in productivity growth μ and a sharp fall in the importance of land in production α after the onset of industrialization. Here we discuss how both of these developments—as well as our small estimate of the elasticity of population growth to real income γ (see later discussion)—contributed to liberating the economy from the Malthusian “iron law of wages,” that is, the notion that wages tend to a very low (subsistence) level. These estimates also reconcile the Malthusian model with episodes historians have identified prior to the Industrial Revolution when some parts of the world have experienced substantial economic growth over a sustained period of time, sometimes several hundred years. Goldstone (2002) refers to these episodes as efflorescences. They include ancient Greece, ancient Rome, Song China, the Islamic golden age, and the golden age of Holland, to name a few. These episodes have often been used as evidence against the Malthusian model (e.g., Persson 2008).

It is important to recognize that steady positive productivity growth in a Malthusian model like ours results in a persistent force pushing wages higher. As wages rise, a counteracting force comes into play pushing wages lower (population growth). The strength of the force pushing wages lower is increasing in the level of the real wage. This implies that as wages rise the downward force gets stronger and stronger and eventually chokes off further increases in wages. In other words, there is a

TABLE V
ESTIMATES OF γ

	Mean	Std. dev.	2.5%	97.5%
Baseline (constant days worked)	0.03	0.05	−0.06	0.12
Variable days worked	0.09	0.02	0.05	0.14

Notes. The table presents the mean, standard deviation, 2.5% quantile, and 97.5% quantile of the posterior distribution we estimate for the elasticity of population growth to income γ . The first row presents results assuming constant days worked (our baseline case), and the second row presents results assuming days worked vary as in [Humphries and Weisdorf \(2019\)](#).

steady-state real wage for each level of average productivity growth in a Malthusian model. This steady state is not at subsistence. Rather, the steady-state real wage is increasing in average productivity growth.

In [Online Appendix I](#), we show that the steady state wage in our Malthusian model is given by

$$(21) \quad \bar{w} = \frac{\mu}{\alpha\gamma} + \text{constant}.$$

As we discussed already, faster productivity growth μ results in a stronger force pushing wages up and therefore a higher steady-state wage. The strength of the counteracting force—which we call the Malthusian population force—is governed by two parameters in our model: the importance of land in production α and the elasticity of population growth with respect to per capita income γ . Intuitively, α determines the slope of the long-run labor demand curve, that is, how rapidly real wages fall as population rises, while γ determines how rapidly the population increases when real wages are high.

With zero productivity growth, the steady-state real wage is potentially very low. Its level depends on factors outside of the scope of our analysis, such as hygiene, the marriage rate, contraceptive technology, and the level of violence in society. With positive productivity growth, the steady-state real wage can be much higher. Whether it is depends on the size of α and γ . We discussed our estimates of α in [Section IV](#). We turn to our estimates of γ .

V.A. The Responsiveness of Population Growth to Income

[Table V](#) presents our estimate of the elasticity of (gross) population growth—that is, $\frac{N_{t+1}}{N_t}$ —with respect to real income γ for our baseline case and the case with variable days worked. In our baseline case, our estimate of γ is extremely small at 0.03. This

means that a 100 log point increase in real wages increases population growth per decade by only 3 log points. Between 1270 and 1440, real wages in England rose by 161% (or 96 log points). This increase in real wages stimulated population growth by a mere 3 log points per decade.

Table V also presents results on γ for our variable days worked case. In this case, we estimate a somewhat larger γ of 0.09. The lower bound of the credible interval in this case exceeds zero. While this estimate is somewhat larger, it remains very small. Allowing for changes in days worked, real incomes in England rose by 70% between 1270 and 1440. A γ of 0.09 implies that this stimulated population growth by 5 percentage points per decade, while a doubling of real income would stimulate population growth by 6 percentage points per decade.

Another way to gauge the quantitative magnitude of our estimates of γ is to calculate the half-life of population dynamics after an exogenous shock to the level of the population, for example, due to a plague. Assuming for simplicity that days worked and the return on capital are constant, that all shocks are equal to their average value, and that there's no productivity growth ($\mu = 0$), we show in Online Appendix I that the dynamics of the population after an initial disturbance are given by the following AR(1) process:

$$(22) \quad n_{t+1} = \left(1 - \frac{\gamma\alpha}{1-\beta}\right) n_t + \text{constant}.$$

The speed of population recovery after a plague-induced decrease is thus governed by $1 - \frac{\gamma\alpha}{1-\beta}$ in this case. In particular, the half-life of the population dynamics (the time it takes for the population to recover half of the way back to steady state after a plague-induced drop) is $\frac{\log 0.5}{\log(1 - \frac{\gamma\alpha}{1-\beta})}$. (The half-life of real wage dynamics is the same.)

Plugging our estimates of the parameters γ , α , and β into the formula, we find that the half-life of population and real wage dynamics before the Industrial Revolution for our baseline case is 356 years. For the variable-days case, the half-life is 115 years. Since α falls sharply after 1760 and β changed little, the Malthusian population force becomes even weaker after 1760. By 1860, the half-life of population and real wage dynamics have risen to 560 years in the baseline case and 240 years with variable days. These long half-lives imply that the strength of the Malthusian population force was weak in England over our sample period—a

result that is sometimes referred to as weak homeostasis. Prior work has also found weak homeostasis. [Lee and Anderson \(2002\)](#) find a half-life of 107 years, and [Crafts and Mills \(2009\)](#) find a half-life of 431 years.

One possible reason we estimate a small value for γ is that the strength of the positive and preventive checks emphasized by Malthus may not have been very sensitive to modest changes in real income. Another possible reason is the quantity-quality trade-off emphasized by modern scholars: higher income may have mostly increased child quality rather than child quantity. Though we estimate a positive γ for our sample, the relationship between real income and fertility switched to being negative in the late nineteenth century and twentieth centuries. This switch may have been due to increases in the value of education tilting behavior even more toward child quality and to increases in the bargaining power of women in marriages.

V.B. Prosperity and the Slope of the Long-Run Labor Demand Curve

We can now use [equation \(21\)](#) and our parameter estimates to assess how changes in the economy after the onset of growth in 1600 affected the long-run steady-state real wage the economy was tending toward. [Figure XIV](#) plots the steady-state wage in our Malthusian economy for different values of productivity growth μ and the slope of the long-run labor demand curve α relative to the steady-state wage with zero productivity growth. Each line in the figure gives the steady-state wage for a particular value of productivity growth as the value of α varies. We do this analysis for the variable-days case, that is, using $\gamma = 0.09$.

The figure illustrates clearly how important the fall in α is for liberating the economy from the iron law of wages. Consider first how steady-state real wages respond to productivity growth when α is equal to our preindustrial estimate of 0.49. In this case, productivity growth of 3% per decade raises the real wage by a factor of two in the long run, while 6% productivity growth raises the real wage by a factor of four. These results show that our Malthusian model is consistent with substantial, multi-hundred year efflorescences of the kind discussed by [Goldstone \(2002\)](#) if we allow for modest productivity growth. The small value we estimate for γ and the associated weak homeostasis are key to this result.

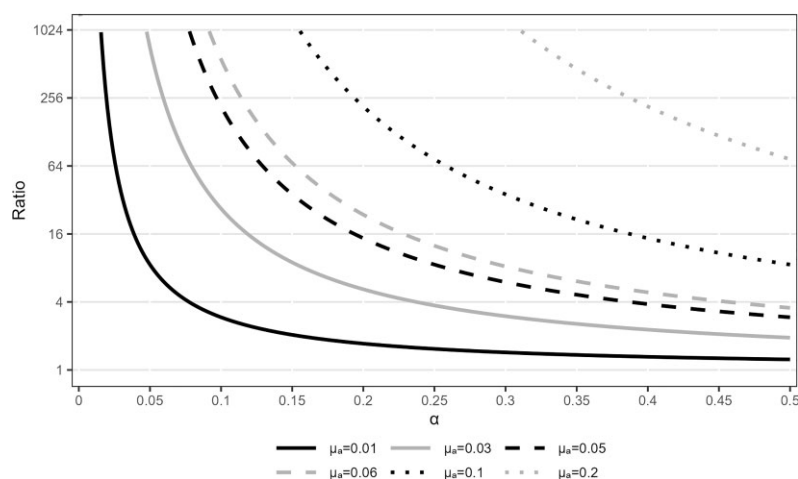


FIGURE XIV

Steady-State Wage with Productivity Growth

The figure plots the steady-state real wage for different average levels of productivity growth (μ) and output elasticity of land (α) relative to the steady-state real wage with zero productivity growth. The parameter α varies on the x -axis and each line plots real wages for a given level of average productivity growth. These impulse responses are calculated assuming that all other model parameters are at their posterior mean values. We do this for the variable-days case.

As α falls, the steady-state real wage for any given level of average productivity growth rises sharply. With our estimate of α for 1860 ($\alpha = 0.22$), productivity growth of 6% per decade can raise the real wage by a factor of 18 in the long run. In other words, this level of productivity growth would have eventually led to a 18-fold increase in real wages even if the demographic transition had not occurred and the Malthusian population force had continued at its 1860 strength. Clearly, a flattening of the long-run labor demand curve is a powerful force for liberating the economy from the iron law of wages when productivity growth is positive.

Another way to visualize these effects is to plot the steady-state wage relative to the actual wage over time. We do this in [Figure XV](#). Before 1600, the ratio of the steady-state wage to the actual wage is relatively stable around one. Once productivity growth begins, the steady-state wage jumps higher and the actual wage only gradually catches up. After 1760, the steady-state wage begins to rise rapidly as the slope of the long-run labor

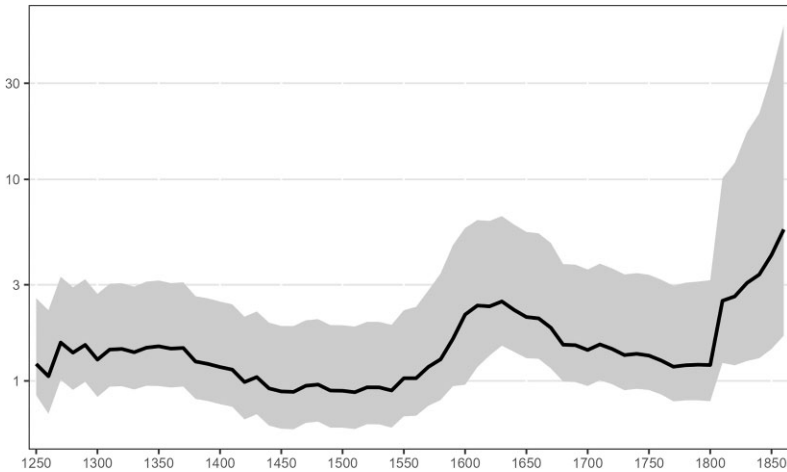


FIGURE XV

Steady-State Wage Relative to Actual Wage

The figure plots the ratio of the steady-state wage—given by equation (49) in the [Online Appendix](#)—and the actual wage over time. Note that the steady-state wage is a function of days worked. In the figure, we use the days worked at each date as the d^* in equation (49). The black line is the median of the posterior for each period, and the gray shaded area is the 90% central posterior interval. We do this for the variable-days case.

demand curve flattens. By 1860, the steady-state wage is more than five times higher than the actual wage.

V.C. Post-1750 Population Explosion

The modest strength of the Malthusian population force in our model raises the question of whether our model can explain, with these parameter values, the large increase in the population of England that occurred after 1750 (see [Figure IV](#)). In 1740, the population of England was 6 million. By 1860, it had risen to almost 20 million. The population grew at a compound rate of 10.4% per decade over this 120-year period.

[Figure XVI](#) compares the evolution of the population in England from 1750 to 1860 with the predicted evolution of the population from the variable-days version of our model. We construct the predicted evolution by taking the evolution of real wages and days worked in England as given and simulating the evolution of the population using [equation \(9\)](#) starting from its actual value in 1740 and assuming no population shocks. This analysis shows that in fact our model with variable days worked

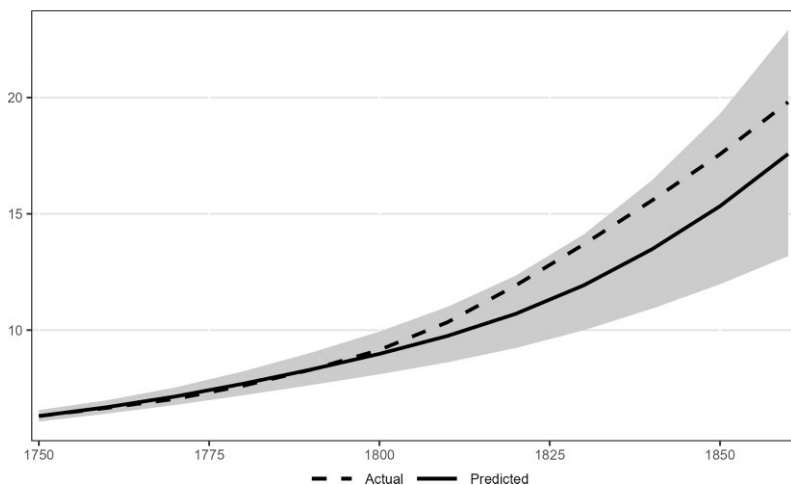


FIGURE XVI

Actual and Predicted Population Dynamics after 1750

The dashed line is the evolution of the population in England. The solid line is the predicted evolution of the population in England from our Malthusian model. In calculating this line, we take the evolution of real wages and days worked in England as given and simulate the evolution of the population using [equation \(9\)](#) starting from its actual value in 1740. The gray shaded area is the 90% central predictive interval given our estimates of α and γ .

can explain the vast majority of the rapid increase in the population between 1740 and 1860. This may seem surprising given the weak Malthusian population force and the somewhat modest increase in real wages over this period. However, allowing for variable days worked, per capita income in England over this period rose quite substantially (see [Figure XII](#)).

VI. PLAGUES AND THE POPULATION

[Figure XVII](#) plots our baseline estimate of the evolution of the population of England from 1250 to 1550 along with prior estimates from [Clark \(2007a\)](#), [Clark \(2010\)](#), and [Broadberry et al. \(2015\)](#). Our estimates are very similar to Clark's. This implies that our estimation procedure largely validates the assumptions Clark makes regarding the evolution of productivity in constructing his population estimates. The estimates of [Broadberry et al. \(2015\)](#) are substantially lower early in the sample period, but then gradually converge.

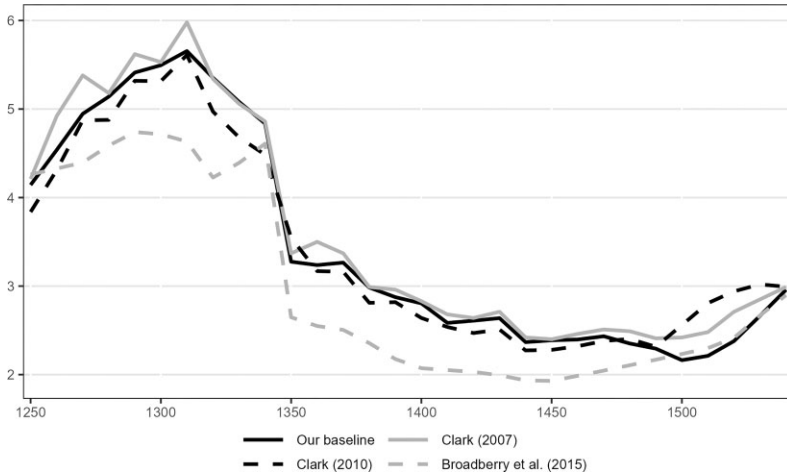


FIGURE XVII

Comparison of Population Estimates for England

The figure plots our estimates of the evolution of the population of England along with estimates from [Clark \(2007a\)](#), [Clark \(2010\)](#), and [Broadberry et al. \(2015\)](#).

The evolution of the population in England over our sample period is heavily affected by plagues. Our model captures plagues (and other influences on the population other than changes in real income) through the shocks ξ_{1t} and ξ_{2t} . [Online Appendix Figure A.9](#) plots the evolution of the sum of these population shocks over the sample period. The largest population shock by far is the Black Death of 1348. We estimate that the population shocks associated with the Black Death led the population of England to shrink by 32%. But [Online Appendix Figure A.9](#) also makes clear that England faced steady population headwinds—persistent negative population shocks—from the early fourteenth century until about 1500.

VII. CONCLUSION

We estimate the evolution of productivity in England from 1250 to 1870 as shifts in the labor demand curve. Our principal finding is that productivity growth began in 1600. Before 1600, productivity growth was zero. We estimate a growth rate of productivity of 2% per decade between 1600 and 1800 and an in-

crease to 5% per decade between 1810 and 1870. Our results indicate that sustained growth in productivity began well before the Glorious Revolution and Industrial Revolution. We demonstrate that the early seventeenth century was a crucial turning point for productivity growth in England, a result that helps distinguish between competing lines of thought regarding the ultimate causes of the emergence of growth.

We attribute the high output growth of the Industrial Revolution only partly to productivity growth. A second important factor was the rapidly falling importance of land in production associated with the transition to steam power fueled by coal. We leverage our model to estimate the strength of the Malthusian population force in preindustrial England. This force was quite weak. The half-life of the response of real wages after a plague-induced decrease in the population was more than 100 years before the onset of the Industrial Revolution and increased to several hundred years by 1860.

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SUPPLEMENTARY MATERIAL

An Online Appendix for this article can be found at *The Quarterly Journal of Economics* online.

DATA AVAILABILITY

The data underlying this article are available in the Harvard Dataverse, <https://doi.org/10.7910/DVN/5EXFLU> (Bouscasse, Nakamura, and Steinsson 2024).

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